



**NATIONAL AUTONOMOUS
UNIVERSITY OF MEXICO
FACULTY OF ENGINEERING**



Preserving Xochimilco: An Interactive Journey

Subject:	Computer Graphics and Human-Computer Interaction
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EVIDENCE SHEET

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SUMMARY

SUMMARY IN SPANISH

This document presents an outline of our final project for the Computer Graphics and Human-Computer Interaction course. This project will focus on developing a graphics pipeline to represent the Xochimilco Ecological Park. The project aims to raise awareness about ecological conservation, chinampa restoration, and the protection of endemic species, using an interactive virtual environment that allows users to explore and understand the challenges facing this unique ecosystem.

In subsequent pages, we will describe the idea behind this project, the objectives to be achieved, and a conceptualization of the models to be implemented. Likewise, we include in this first installment the software and hardware requirements necessary to run the virtual environment. Finally, we add our conclusions from the beginning of the project, as well as the expected lessons learned upon its conclusion.

SUMMARY IN ENGLISH

This document aims to present an outline of our final project for the Computer Graphics and Human-Computer Interaction course, which focuses on developing a graphics pipeline to represent the Xochimilco Ecological Park.

In the following pages, we will describe the project's concept, objectives, and an initial design of the models to be implemented. Additionally, this first submission includes the software and hardware requirements necessary to run the virtual environment. Finally, we provide our initial conclusions as we begin the project, along with the expected learnings upon its completion.

INTRODUCTION

The Xochimilco Ecological Park, located south of Mexico City, is a nature reserve that is part of the "Ejidos de Xochimilco y San Gregorio Atlapulco" Protected Natural Area and was declared a World Heritage Site on November 11, 1987 (UNESCO, 1987). Since its creation in the late 1980s, its main objective was to create a space that would allow for the ecological conservation and restoration of the chinampero ecosystem, a pre-Hispanic agricultural technique that has been key to the economy and identity of Xochimilco (Rojas Villamar et al., 2020).

Over the years, from the completion of its construction in 1993 until its reopening in 2020, it has served as a refuge for various species of flora and fauna; such as resident and migratory birds, insects, rodents, as well as the home of the axolotl, an amphibian endemic to the lakes of Xochimilco and in danger of extinction, recognized mainly for its regenerative capacities. Therefore, it has allowed to promote environmental education while offering recreational spaces for visitors; such as a collective market area, spaces for sports activities, cycling circuits, docks that allow connection to the chinampas or the Cuemanco pier, as well as the "Flor de Chinampas Museum" or also known as "Chinampaxóchitl", which includes exhibitions of a wide variety of species belonging to the region. (Rojas Villamar et al., 2020).

Having discussed the importance of this sustainable project, it is necessary to emphasize the ecological challenges it faces, such as water pollution in lakes and canals due to sewage and household waste infiltration, which affects the local flora and fauna. Therefore, through our project, we seek to raise awareness about the importance of this ecological space, promoting environmental education, the rehabilitation of traditional chinampas, and sustainable ecotourism.

State of the Art:

The use of advanced graphic technologies in environmental education has grown considerably in recent years, especially in the context of urban and natural ecosystem conservation. Interactive virtual environments, in particular, have become powerful tools for raising public awareness about issues of ecology, sustainability, and environmental protection.

In the case of the Xochimilco Ecological Park, various investigations have highlighted the importance of this natural space, which, as part of the Ejidos de Xochimilco and San Gregorio Atlapulco Natural Protected Area, is home to a high level of biodiversity, including endemic species such as the axolotl and traditional chinampas, a pre-Hispanic agricultural system that was key to the local economy and culture (UNESCO, 1987). However, this park faces serious ecological challenges, mainly due to water pollution in the canals, which has been caused by sewage infiltration and domestic waste (Rojas Villamar et al., 2020).

Regarding the use of graphic technologies to represent ecological spaces, recent studies have shown that 3D visualization and interactive simulations are effective in demonstrating ecological restoration processes, such as the rehabilitation of chinampas and biodiversity protection. For example, in other urban ecosystems such as national parks or coastal areas, 3D models have been used to simulate the restoration of damaged habitats, providing a means for users to interact with the environment and understand the impact of their actions on the environment (Baker et al., 2019; Ferreira et al., 2018). These tools not only allow the visualization of ecosystem evolution but also promote environmental education by offering users an immersive and direct experience with the problem.

The Xochimilco project draws on these technological approaches, using OpenGL and other graphic resources to create an interactive platform that

allows users to explore the park and learn about the ecological threats it faces, such as canal pollution and chinampa degradation. Furthermore, this approach also seeks to promote sustainable ecotourism by educating visitors about the importance of conserving the park and its biodiversity. According to studies by García et al. (2020), the implementation of interactive technologies in eco-tourism areas has the potential to improve users' environmental awareness and encourage responsible tourism.

Therefore, it is also worth highlighting the importance of the Xochimilco Ecological Park not only as an ecological reserve but also as a tourist destination where the population can go to learn about the unique ecosystem found in the area and the ongoing efforts aimed at preserving this area.

Computer graphics has the potential to model different sites with the intention of developing tourism in a certain area. According to a study by Liu & Quan. (2022), 3D modeling technologies have been used to model tourist and ecological reserve sites such as the Huiyan Peak Scenic Site in China. Technologies like these make it possible to revitalize and strengthen the tourism industry, especially in the post-pandemic era, which has as its main focus the recovery of said industry, which was one of the most affected by this global event. Therefore, a graphic application of this style can motivate tourists to visit the Xochimilco ecological park responsibly, safeguarding the well-being of the local ecosystem.

On the other hand, chinampa rehabilitation has become a key strategy not only for the conservation of this ecosystem, but also as a tool for environmental education and awareness. Research such as that by Arenas et al. (2017) has shown that the restoration of chinampa systems in Xochimilco not only has ecological benefits but also social benefits, by involving local communities in the process of restoration and management of the park.

Therefore, the state of the art shows that interactive graphic technologies and 3D simulations can play a fundamental role in promoting ecological conservation and tourism in urban areas like Xochimilco. By combining interactive visualization, environmental education, and ecotourism promotion, the aim is to offer a comprehensive platform that not only educates users about Xochimilco's ecological problems but also encourages them to actively participate in its conservation.

PROBLEM STATEMENT

Today, Xochimilco is widely recognized for its culture, traditions and, in this regard, for its emblematic canals and chinampas, which have faced serious pollution problems for decades that threaten their ecosystem and cultural heritage. According to Excélsior (2023), there have been warnings for years about the risk of their disappearance, mainly due to the extraction of groundwater in the capital. Therefore, efforts have been made to mitigate the drying up by introducing recycled water, which, although it has helped maintain water levels, has led to the introduction of bacteria and heavy metals, in addition to other contaminants.

According to the article entitled “Environmental deterioration and irregular urban occupation in the Lake Zone of Xochimilco, Mexico City” (Rojas Villamar, I. et al., 2020). The process of peri-urbanization in Mexico City, this has led to irregular land occupation, closely linked to environmental degradation due to the discharge of gray and sewage into the canals. This article demonstrates that illegal settlements, in collusion with local authorities who tolerate such irregularities, pose a risk to Xochimilco's status as a World Heritage Site.

In addition to the problems of pollution and water extraction, the lack of an effective conservation strategy has exacerbated the degradation of Xochimilco's ecosystems. In particular, the chinampas, a traditional agricultural system that has been crucial to the local identity and economy, continue to suffer from unsettled urbanization and pollution. Despite restoration efforts, these have

been limited in effectiveness, highlighting the need for an innovative approach to promote their conservation. In this context, our project seeks to use interactive graphic technologies to raise awareness about these ecological issues, showing how the rehabilitation of the chinampas and the protection of biodiversity can contribute to the park's sustainability. Through a virtual environment, we aim to raise awareness about the importance of Xochimilco, promoting both environmental education and sustainable ecotourism, which is key to its long-term preservation.

Political, Ethical and Social Aspects:

The political aspect of preserving the Xochimilco Ecological Park is marked by the interaction between local authorities and the communities that live in the area. Over the years, irregular land occupation and the authorities' tolerance of illegal settlements has complicated the implementation of effective policies for the park's conservation (Rojas Villamar et al., 2020). This lack of land use control not only affects biodiversity but also jeopardizes Xochimilco's status as a World Heritage Site. Restoration efforts are hampered by the lack of a comprehensive policy that engages both residents and policymakers in the preservation process.

Regarding the ethical aspect, one of the main dilemmas is the balance between environmental conservation and the needs of local communities. Ecological restoration projects must ensure that the solutions implemented are not only environmentally sustainable, but also respect the rights and quality of life of the people who depend on the Xochimilco ecosystem for their livelihoods (García et al., 2020). Ethics in natural resource management requires that decisions consider not only conservation, but also the well-being of the people who live in these vulnerable areas.

On the social side, the impact of pollution and ecosystem degradation directly affects the communities of Xochimilco, as many families depend on the area's

natural resources for agricultural and tourism activities. The decline in water quality and loss of biodiversity impact both the local economy and the region's cultural identity, as chinampas and endemic species are key elements of daily life and the local economy (Rojas Villamar et al., 2020). Ecotourism, which could be an important source of income, is limited by the lack of adequate infrastructure and the perception that the area is polluted.

Problem statistics:

According to Excélsior (2023), groundwater extraction in Mexico City has reduced water levels in Xochimilco's canals by more than 30%, intensifying the drying out of the chinampas. Despite efforts to introduce recycled water, this measure has generated serious side effects such as the presence of bacteria and heavy metals in the canals. A study by Rojas Villamar et al. (2020) highlights that more than 60% of settlements in Xochimilco are illegal, directly contributing to pollution and excessive land occupation.

In addition, the Xochimilco Ecological Park faces several challenges, among which the serious situation of the axolotl stands out, whose population has decreased enormously, going from 6,000 individuals per square kilometer in 1998 to only 36 in 2014. This is largely due to the contamination of the canals, the presence of invasive species such as tilapia, and uncontrolled urbanization that has destroyed its natural habitat (El País, 2024).

According to studies, more than 60% of the settlements in the area are illegal, which has exacerbated wastewater discharge and the uncontrolled growth of urbanization in protected areas (Rojas Villamar et al., 2020). This peri-urbanization process has also been responsible for deforestation, as more than 60% of the area is severely deforested, affecting biodiversity and essential ecosystem services (Wikipedia, 2024).

Furthermore, excessive groundwater extraction has caused the area to subside by up to 18 cm per year, affecting infrastructure and canals

(Wikipedia, 2024). In terms of agricultural production, producers of cempasúchil, a traditional flower used on the Day of the Dead, face difficulties due to flooding and deteriorating roads in the region, which impacts their production (El País, 2024). These ecological, social, and economic problems underscore the importance of implementing restoration and environmental education projects, such as the development of interactive virtual environments, to promote awareness and conservation of the Xochimilco ecosystem.

Relationship with the UN 2030 Agenda for Sustainable Development Goals:

The Xochimilco Ecological Park and its conservation process are closely linked to the goals of the 2030 Agenda for Sustainable Development adopted by the UN General Assembly. In particular, the environmental and social challenges facing Xochimilco reflect several of the Sustainable Development Goals, such as Goal 6: Clean Water and Sanitation, which seeks to ensure the availability and sustainable management of water and sanitation for all. Pollution of Xochimilco's canals due to wastewater and groundwater extraction threaten water quality, preventing this goal from being achieved in the region.

Goal 11: Sustainable Cities and Communities, which promotes inclusiveness, sustainability, and resilience, is also at stake in Xochimilco. Uncontrolled urbanization and illegal settlements have contributed to the deterioration of this unique ecosystem. An integrated approach is needed, involving both ecological restoration and responsible urban planning that respects the area's natural resources.

Likewise, Goal 14: Life Below Water highlights the need to conserve and sustainably use oceans, marine life, and marine resources, which is directly applicable to the conservation of the Xochimilco canals, which are home to endemic species like the axolotl. Protecting biodiversity and restoring aquatic habitats are essential to meeting this global goal.

Finally, Goal 15: Life on Land emphasizes the need to manage and restore terrestrial ecosystems. Restoring Xochimilco's traditional chinampas is crucial not only for their aquatic ecosystem but also for maintaining an ecological balance that benefits both local biodiversity and the communities that depend on this agricultural system.

JUSTIFICATION

The importance of the Xochimilco Ecological Park as part of a project for the preservation of local flora and fauna has been mentioned. We have highlighted the ecological challenges it has presented. However, the future of this project lies in the effective implementation of strategies in collaboration with local authorities, local communities, and environmental organizations, so that it meets the objectives for which it was developed. That said, the initial idea of our project is to carry out the development of a virtual environment that will allow us to raise awareness about this ecological project, as well as promote environmental education through the rehabilitation of traditional chinampas, teaching about the species that inhabit this region and are threatened by the aforementioned conditions, and promoting ecotourism. All of this is key to ensuring the preservation of this invaluable heritage for future generations.

Finding a solution to the problem at hand is essential due to the unique importance of Xochimilco as an urban ecosystem that not only represents a cultural and natural symbol of Mexico City but also harbors irreplaceable biodiversity. The pollution of its canals, the degradation of its chinampas, and the threat of extinction of species like the axolotl have a direct impact on the region's environmental health and the well-being of local communities. The environmental crisis facing Xochimilco not only endangers its natural resources but also its potential to become a model for conservation and sustainable ecotourism.

Furthermore, the lack of awareness about ecological problems and possible solutions hinders the necessary collaboration between authorities, communities,

and key stakeholders in the restoration process. Through the creation of an interactive virtual environment, this project seeks to raise public awareness about the urgency of conservation, promoting greater commitment to ecological restoration and responsible ecotourism. This innovative approach has the potential to engage the community, tourists, and authorities in a collective effort to ensure the park's sustainability, thus contributing to the preservation of Xochimilco as a world heritage site for future generations. The importance of a solution lies in the fact that Xochimilco is not just a tourist attraction, but an invaluable resource that must be protected to maintain its biodiversity, its history, and its function as a unique ecosystem.

GENERAL OBJECTIVE

The general objective of this project is to create a graphic application whose environment represents the ecological park of the Xochimilco delegation in Mexico City using different geometric modeling techniques to represent the different buildings, trees and different structures that can be found in its real counterpart so that the user can freely move and explore said environment and learn about this park.

SPECIFIC OBJECTIVES

The specific objectives we contemplate for this project are listed below.

- Through project creation, participants will learn to program a desktop application that leverages the advantages of heterogeneous computing and graphics cards to create a virtual environment.
- The user will be able to explore the facilities of the Xochimilco ecological park through a window and will be able to move forward, backward, right, left and look in different directions.

- Through the use of the software, the user will learn about the flora and fauna found in the Xochimilco Ecological Park and the efforts being made to conserve it.
- Using this graphic application, users can learn about the importance of protecting the environment and the natural heritage found in the Xochimilco district.

REQUIREMENTS ANALYSIS

Software.

For this project, we are considering using software to carry it out.

We require the use of an IDE with support for the C and C++ programming languages to be able to use OpenGL and related libraries such as assimp for loading models into our virtual environment.

The Visual Studio 2022 IDE is proposed due to its relatively simple handling when writing and compiling programs, and because it is the tool with which the team presenting the project has the most experience due to the practices they have carried out in the Computer Graphics and Human-Computer Interaction laboratory.

The use of Windows 10 or higher is contemplated as the operating system since it is the platform for which we will be developing our application.

We consider using texture editing software such as GIMP or Krita to texturize our models or design user interface elements.

We also consider using version control software like GitHub to keep track of changes made to the project's source code along with the files we use as templates in our environment.

Hardware.

To develop the project, we require at least two computers with Windows 10 or higher and a graphics card, since we will be calculating several geometries in real time when running the application. Therefore, we want to take advantage of heterogeneous computing when rendering images using the graphics pipeline offered by OpenGL.

The second computer will be used to ensure that the program runs on a different computer than the one used for its development.

Human Resources.

The project proposal team is made up of four people who have been involved in the initial planning and documentation of the project.

Following this, the participation of all members in the active development of the graphical application is contemplated in activities such as:

- Modeling missing objects.
- Programming.
- Research information on the park's flora and fauna.
- Quality control.
- Delivery of the executable and final project report.

Services.

We are considering some essential services for the development of this project, primarily electricity to power the computing equipment that will be used to build our graphical application.

Cost table.

For development, we consider using existing personal equipment and free software to optimize costs, although the following cost table can also be taken into account in any case (all prices in Mexican pesos):

Category	Element	Approximate cost (MXN)	Notes
Software	Visual Studio 2022	\$0	Free IDE for development
	OpenGL	\$0	Free graphics API
	Blender	\$0	Free modeling software.
	GitHub	\$0	Version control platform.
	GIMP, Krita	\$0	Free alternatives to Photoshop for creating textures and UI elements.
Hardware	PC for development	\$10912 (\$0 if using existing personal equipment)	You can work with existing equipment. Hypothetical price for a PC with 16GB RAM, Intel i5 3.7GHz CPU, Nvidia GPURTX 3050. (Amazon, 2025)
Human Resources (labour)	Teamwork	\$2800, \$2500, \$4000, \$3500, \$200 and \$1500	Work proposal and object modeling, visual effects such as texturing, programming, animation, executables, and documentation.
Services	Electricity	\$800 - \$1400	Electricity service to power computer equipment

Giving an estimated total for the project of \$26212-\$26812 MXN.

WORK TEAM

Therefore, to carry out this project we have clarified the roles to be covered by our team:

- Project Leader. Responsible for coordinating and reviewing each stage of the project, ensuring that established goals are met on schedule, while properly managing resources and ensuring efficient performance of team members, fulfilling the general and specific objectives described at the beginning.
- 3D Designer. This person is responsible for designing and creating the three-dimensional models that will be represented in the virtual scene, from characters simulating park activities to the ecosystem's flora and fauna.
- Animator. Responsible for implementing animations for characters, flora, fauna, and objects, with the goal of replicating behavior and interactions.
- Visual effects and lighting specialist. He will be responsible for applying lighting models to enhance the immersive experience and give the scene a realistic feel.
- Programmer. Will be responsible for developing the interactive graphics system using OpenGL and C++, in order to implement shading, audio proximity, transformations, etc.
- Test Manager. This person will be responsible for ensuring the system works properly. They will perform functionality and usability tests, as well as code packaging, ensuring the product is delivered as an executable ready for deployment in the graphical system.

METHODOLOGY

Preliminary Design

This document addresses the initial design of our final project, which will represent the Xochimilco Ecological Park. We chose this setting because it will allow us to represent, through models, animations, lighting, and textures, one of the benchmarks for ecological conservation, combining Xochimilco's flora and fauna with sustainable tourism.

Below are the initial sketches of the models we have thought of to represent this place, as well as the sketch of the place.

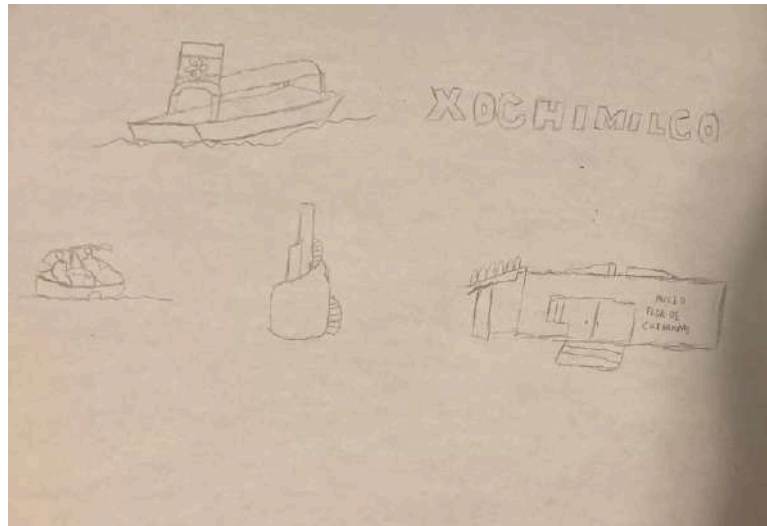


Image 1. Referring to the Initial Sketch of the Project.

In the first sketch, we thought of the following initial models of the project:

- Trajinera. This model will be one of the main objects present on the stage, as some will be stationary and others will be moving on Lake Huetzalín.
- Boat. This other object will be found stationary and moving on Lake Acitlalin.

- Flor de Chinampas Museum. This museum will be part of the buildings we'll be representing on stage, and will also offer a brief tour of its interior.
- Xochimilco. This will be a sign at the park entrance, which will guide us as we begin our tour of the stage.
- Tower. This latest model will serve not only as a decorative element of the stage, but also as a viewing platform for a different perspective of the project.

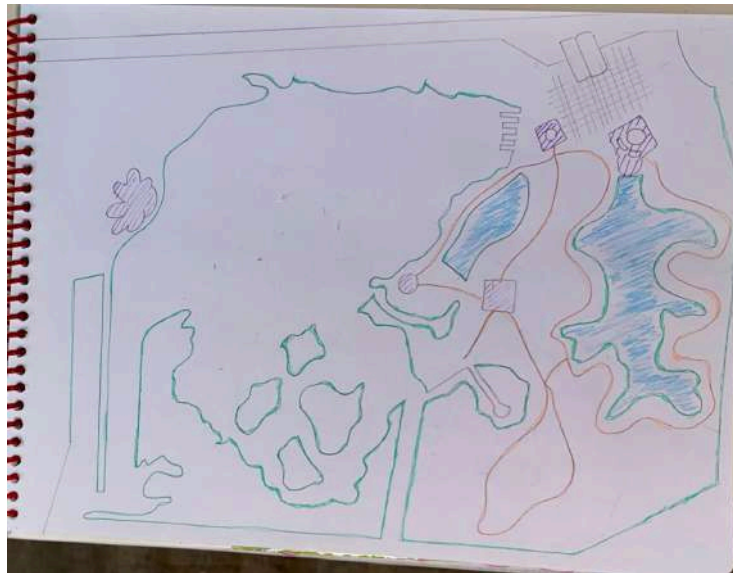


Image 2. Regarding the Initial Sketch of the Project.

The second draft initially includes the Xochimilco Ecological Park, which will be bordered to the north by a car circuit representing the Periférico roadway. The Cuemanco Canal and its pier will be located on the west side. The southern and eastern areas of the project will be bordered by the same park. Two lakes are being considered: Lake Huetzalin (the main lake), which is expected to better represent the local flora and fauna, in addition to featuring moving models of trajineras. The second Lake, Acitlalin, will also feature models of species and boats, but to a lesser extent due to its size. This first draft details five areas of interest, which will be connected by a path that will serve as a guide within the project.

Work Schedule

WORK SCHEDULE		
From: 03/06/2025		To: 05/06/2025
Week	Activity	Objectives
Week 1 (March 6-12)	Definition of the subject, tools and technology research, limitation of the project's reach.	Establish the project's approach, select the necessary tools and define the functionalities of the initial prototype.
Week 2 (March 12-19)	Preliminary interface design and definition of key graphics.	Create a functional user interface, decide the 3D models to be used and VFX for the project.
Week 3 (March 20-26)	Basic prototype development and internal testing.	Implement 3D viewing and verify the correct functioning of interactions and graphics.
Week 4 (March 27-April 2)	User testing and feedback analysis.	Get feedback on the interface and verify the clarity of the ecological message.
Week 5 (April 3-9)	Prototype refinement and iteration testing.	Upgrade the graphics and optimize the user experience based on the feedback.
Week 6 (April 10-16)	Development of the final prototype and complete system testing.	Complete the prototype, ensuring proper functioning and graphical performance
Week 7 (April 17-May 6)	Validation with final users and delivery of the project.	Make final tests, deliver the software, demo video, reports and final presentation.

Image 3. Regarding the Project Schedule.

Week 1 (March 6 - March 12):

This week, the project's theme and focus will be defined, with a specific focus on creating a virtual environment to showcase the efforts to preserve the Xochimilco Ecological Park. This includes the rehabilitation of chinampas, the protection of endemic species, and the promotion of sustainable ecotourism. The necessary tools and technologies, such as OpenGL 3 or higher, GLSL, and 3D modeling software, will be investigated to create an interactive graphical environment representing key elements of the park, such as flora, fauna, and traditional chinampas. The project's scope will also be defined, defining the main functionalities for the first prototype, prioritizing the graphic visualization of biodiversity and the effects of pollution, as well as ecological rehabilitation and restoration actions.

Week 2 (March 13 - March 19)

During this week, the project's preliminary interface will be designed, creating user interface sketches that facilitate navigation through the park's different areas, visualization of endangered species, and the chinampa rehabilitation process. In addition, key graphics will be defined, with decisions made on 3D models of the chinampas, the axolotl, and other endemic species, as well as visual effects to represent canal pollution and ecosystem restoration.

Week 3 (March 20 - March 26)

A basic functional prototype will be developed, implementing the initial 3D visualization with key elements, such as chinampas, endemic species, and a representation of environmental issues, such as water pollution. In addition, internal testing will be conducted to verify that the graphics render correctly and that basic interactions work effectively, such as exploring park areas and visualizing ecological issues.

Week 4 (March 27 - April 2)

Tests were conducted with a small group of users to obtain feedback on the graphical interface, ease of navigation, and clarity of the message regarding environmental education and ecological issues. This feedback will then be analyzed to determine whether the information on ecological rehabilitation and the protection of endemic species is clear and understandable to users.

Week 5 (April 3 - April 9)

This week, the prototype was refined, improving the chinampa graphics and the visual effects of pollution, adjusting the lighting and texturing to adequately reflect the deteriorated state of the canals. Iteration testing will be conducted to fine-tune the graphic elements and interactions based on feedback, improving the user experience and optimizing the representation of ecological rehabilitation.

Week 6 (April 10 - April 16)

The final version of the prototype will be developed, completing all functionalities, including interactive simulations on chinampa restoration and water quality improvement. Comprehensive system testing will also be conducted to ensure adequate graphical performance and user interaction across different platforms.

Week 7 (April 17 - May 6)

Over the past week, validation was conducted with end users, ensuring that the final product meets the requirements for graphical representation and interaction functionality. Finally, the project will be delivered, including the packaged executable file, the demonstration video, reports in Spanish and English, and the final presentation.

Geometric Modeling

Flora and Fauna:

1. Sauces.



Image 4. Willow Modeling.

2. Tepozanes.



Image 5. Modeling of Tepozanes.

3. Pumpkins.

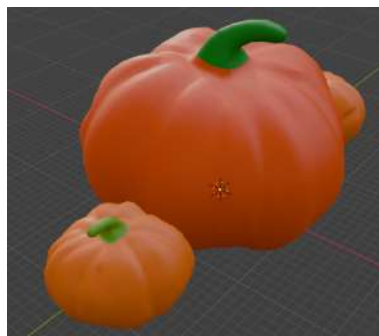


Image 6. Pumpkin Modeling.

4. Vegetable plantation.



Image 7. Modeling of vegetable crops.

5. Flower plantation.



Image 8. Flower planting modeling.

6. Poinsettia flowers.

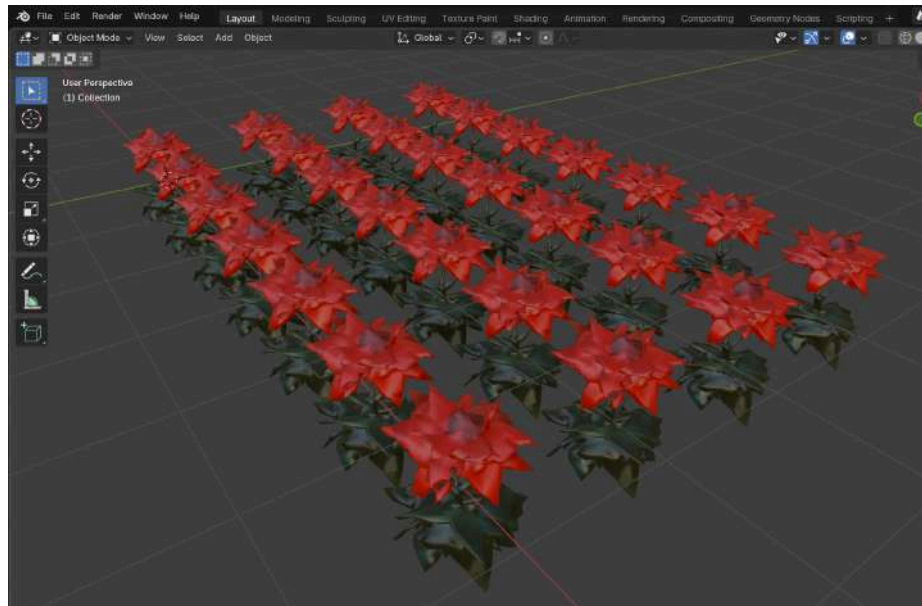


Image 9. Modeling poinsettias.

7. Marigold flowers.



Image 10. Modeling of Marigold Flowers.

8. Flores.

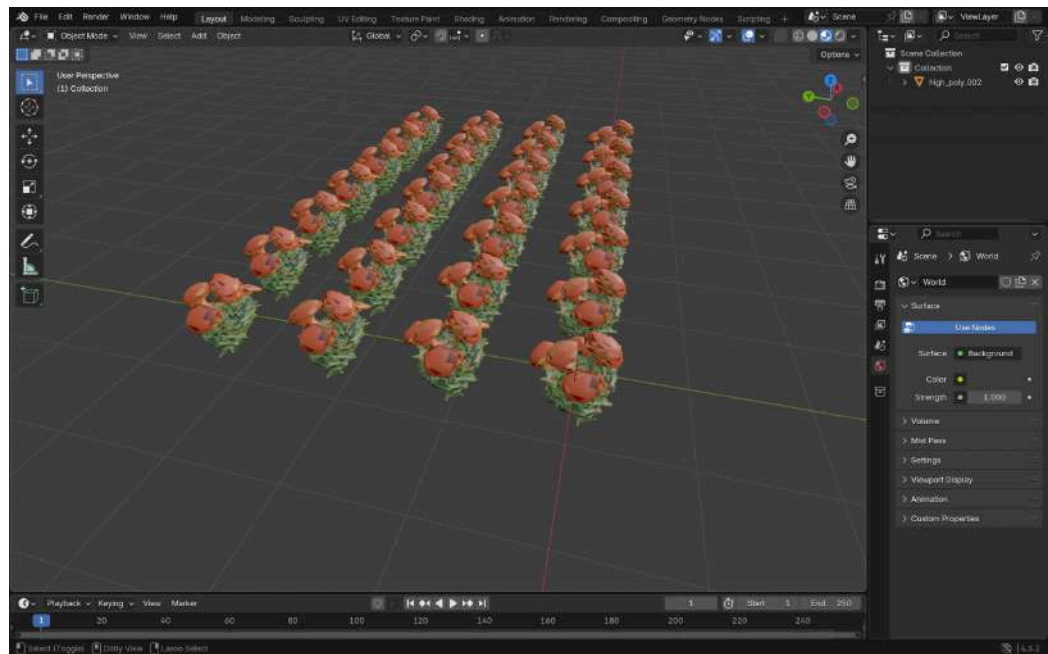


Image 11. Flower modeling.

9. Trees.

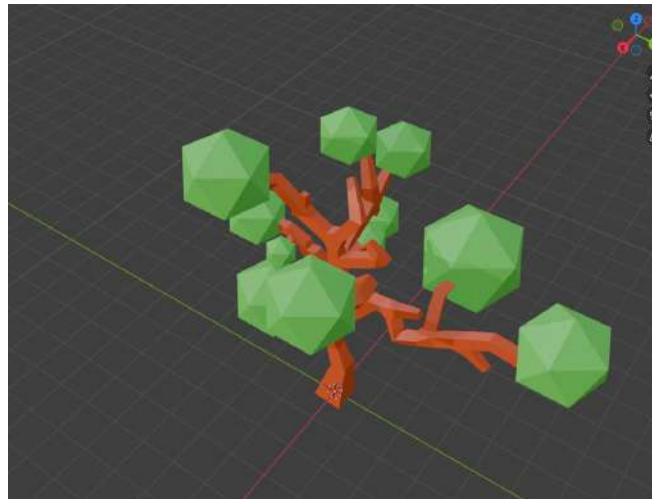


Image 12. Tree Modeling.

10. Sparrows.

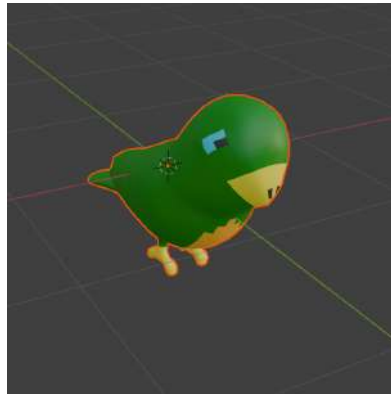


Image 13. Sparrow modeling.

11. Eagles.



Image 14. Modeling an eagle.

Structures:

1. Exercise area.

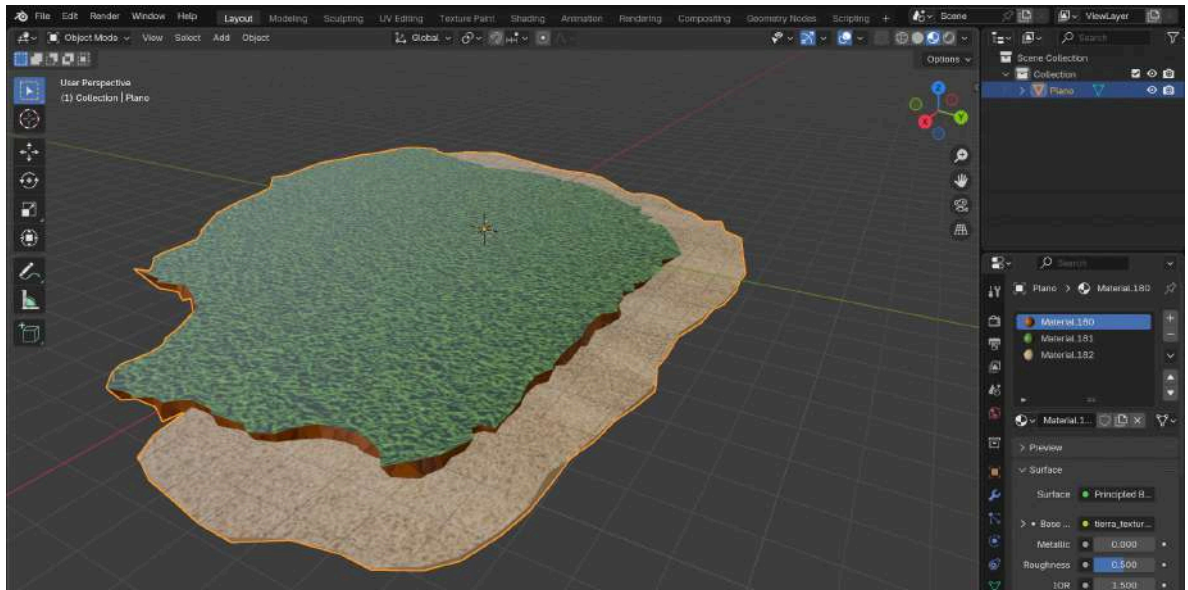


Image 15. Modeling exercise zones.

2. Lagos.

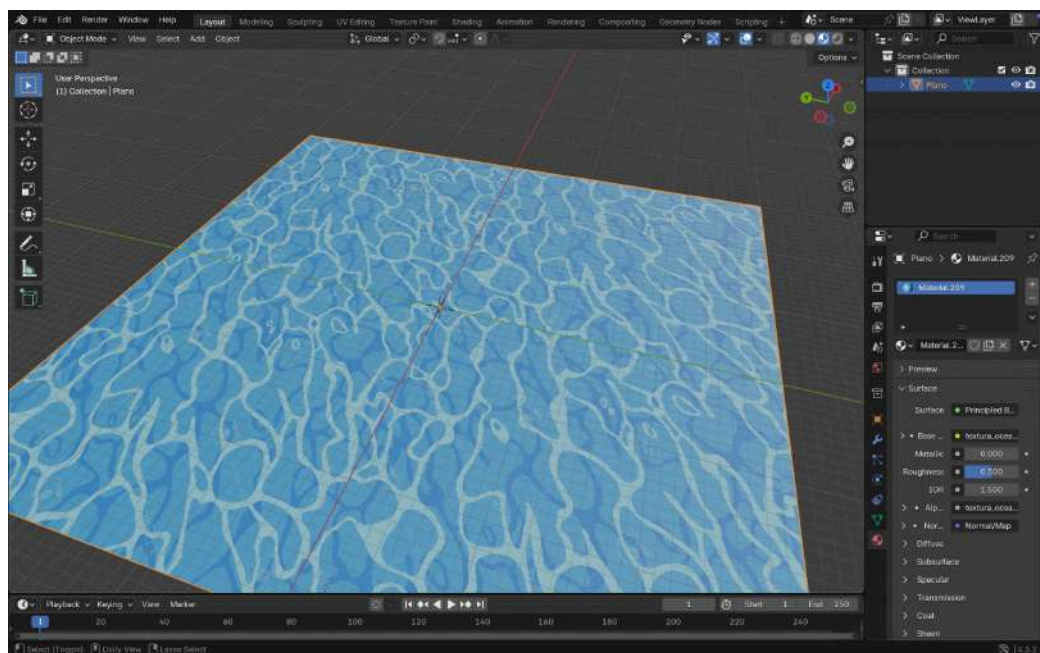


Image 16. Lake Modeling.

3. Irregular settlements.



Image 17. House modeling.

Objects:

1. Dining room tables.

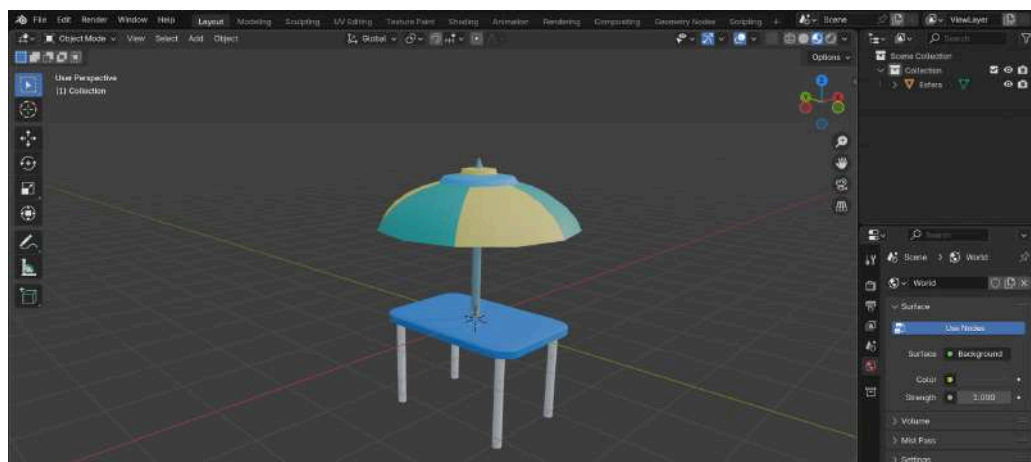


Image 18. Modeling tables for the dining room.

2. Food.

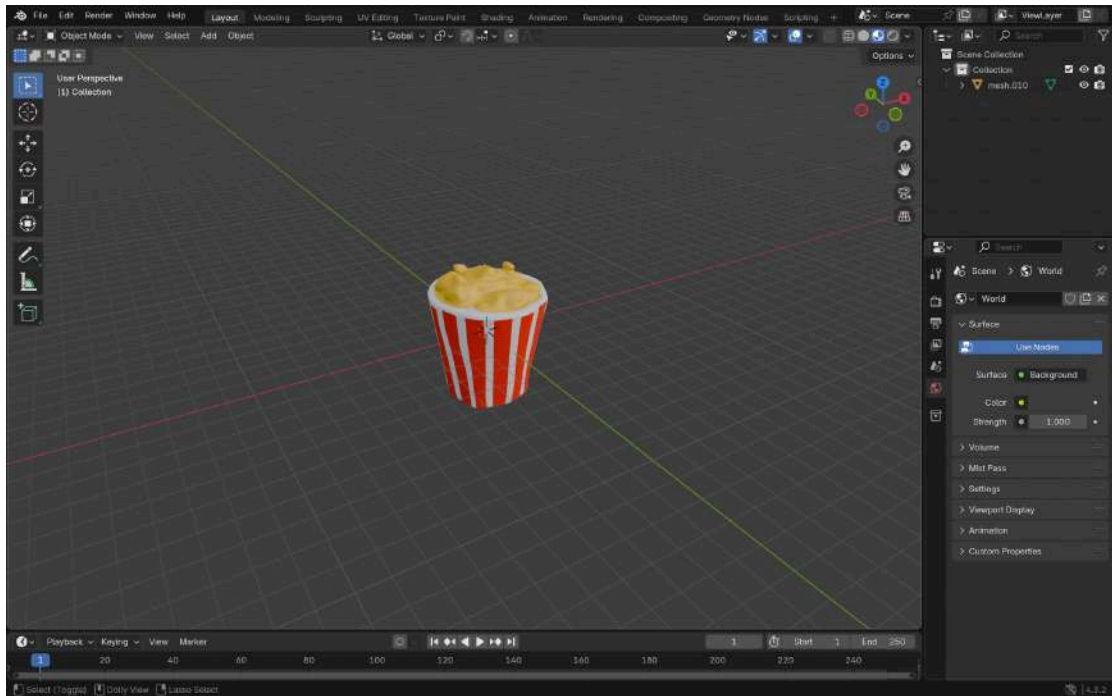


Image 19. Food Modeling.

3. Souvenirs from the museum.

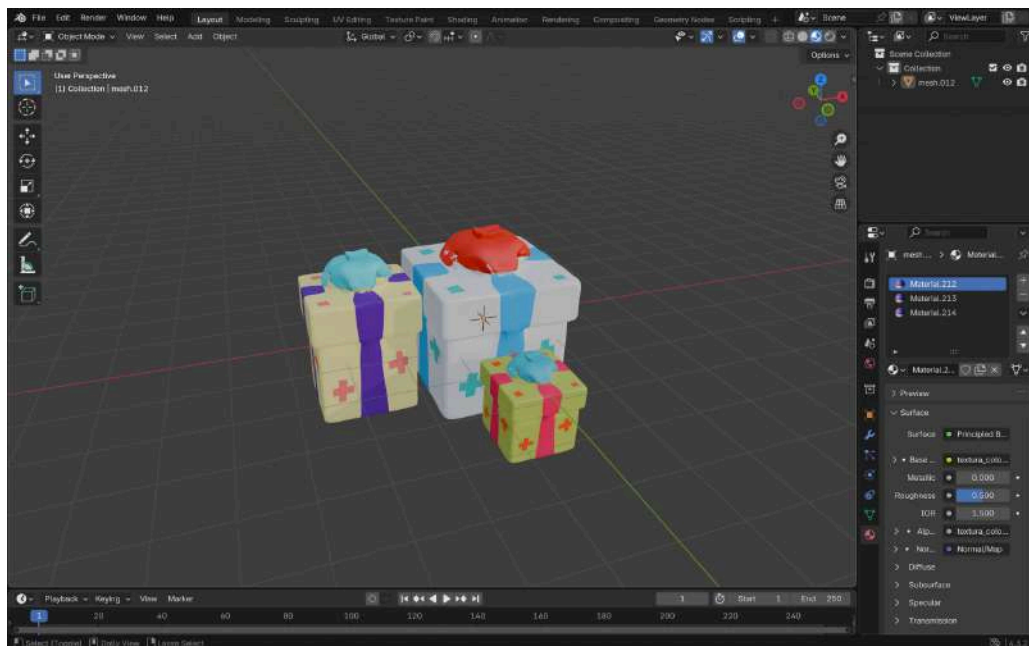


Image 20. Modeling of Museum Souvenirs.

4. Community market products.

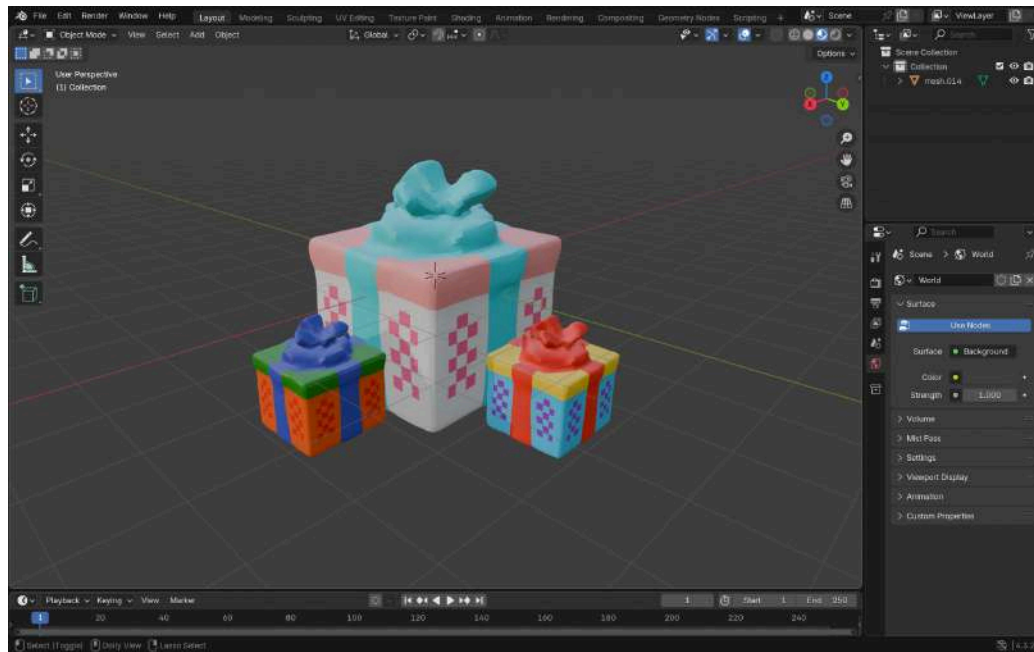


Image 21. Market Product Modeling.

5. Trajineras.



Image 22. Modeling of Trajinera.

6. Decorations.



Image 23. Modeling Decorations.

7. Ice cream cart.



Image 24. Modeling an Ice Cream Cart.

8. Sales stands.



Image 25. Modeling of sales positions.

9. Field tables.

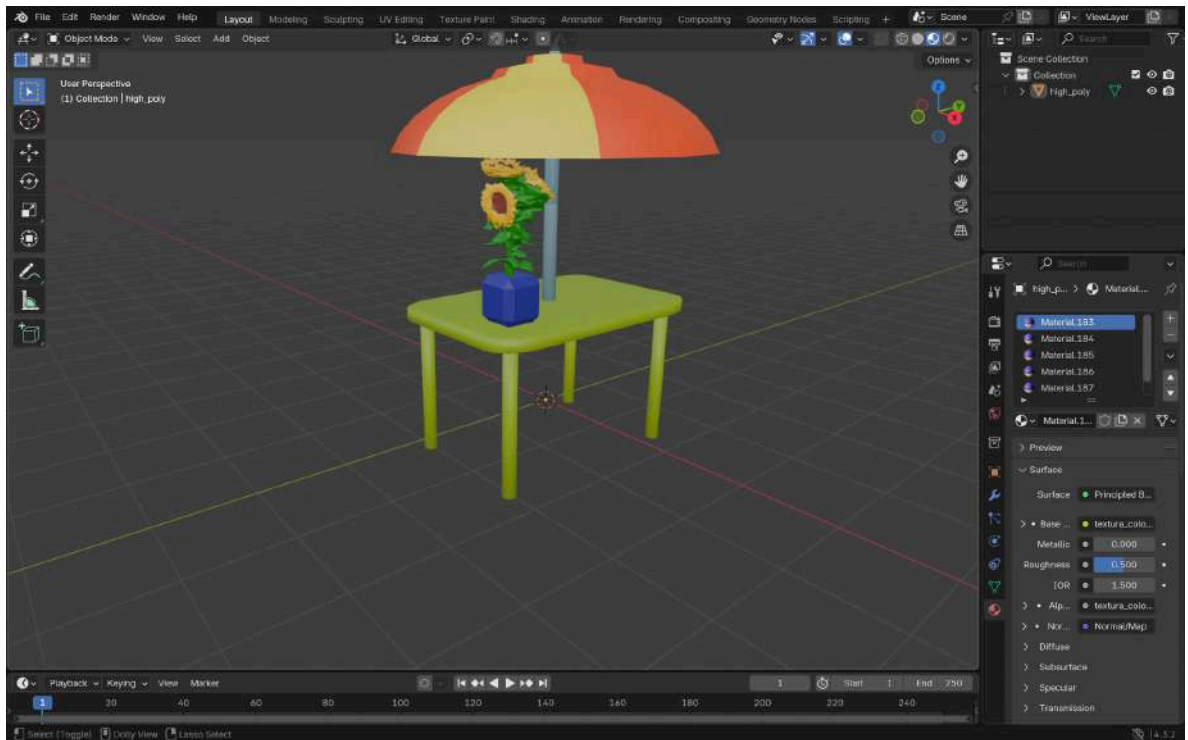


Image 26. Modeling field tables.

10. Stalls.

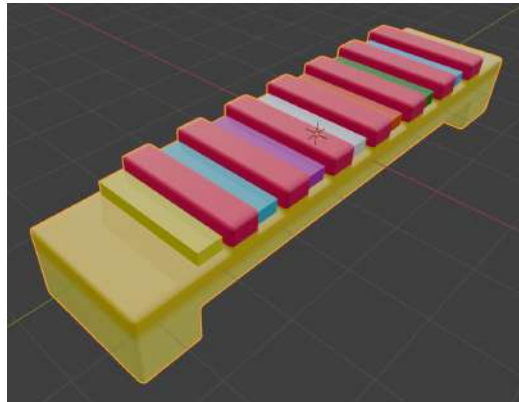


Image 27. Bench Modeling.

11. Reference books.

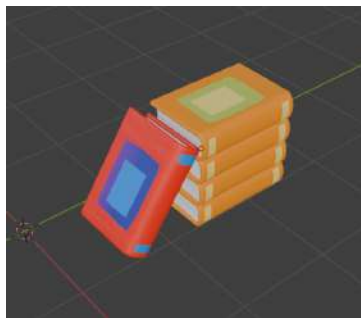


Image 28. Reference book modeling.

12. Museum documents.

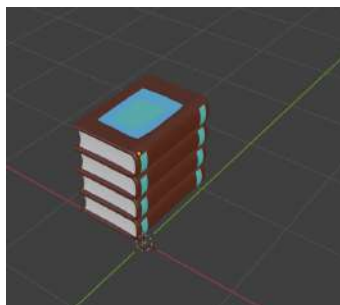


Image 29. Modeling museum documents.

13. Books for visitors.

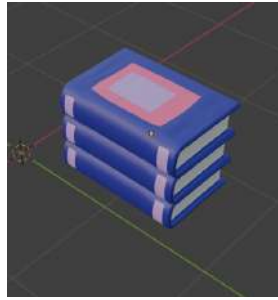


Image 30. Modeling of visitor books.

14. Basketball ball.

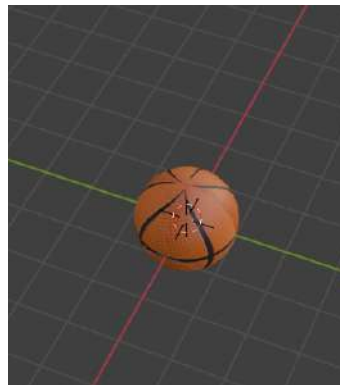


Image 31. Modeling a Basketball.

15. Luminaires.



Image 32. Lantern Modeling.

16. Chinampas.



Image 33. Chinampa modeling.

Museum Objects:

1. Vases.

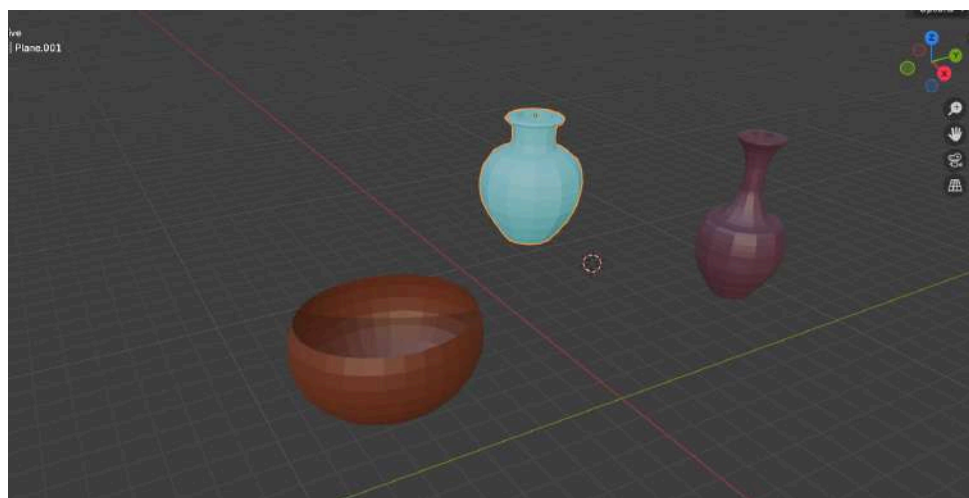


Image 34. Vase modeling.

2. Bank

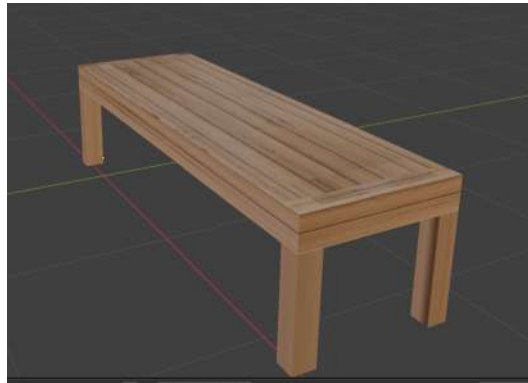


Image 35. Bank modeling.

3. Paintings

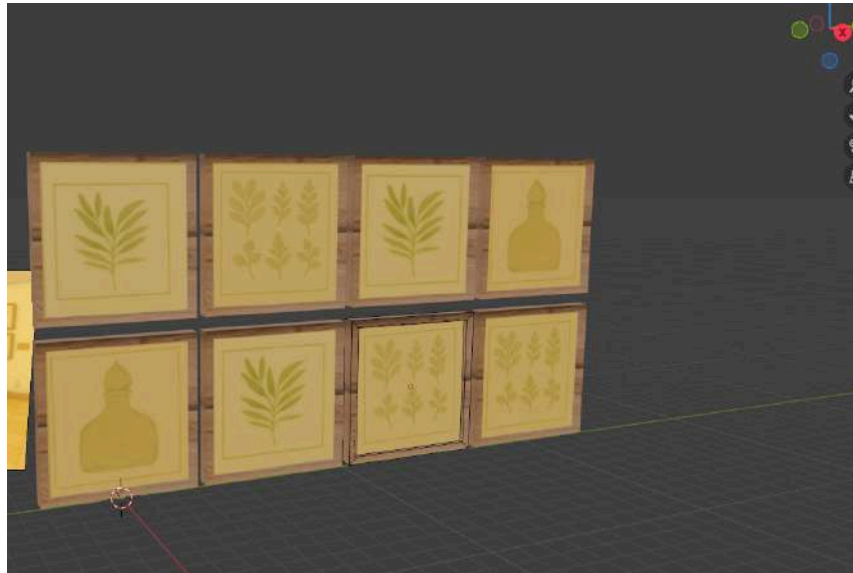


Image 36. Frame modeling.

4. Mini trajinera

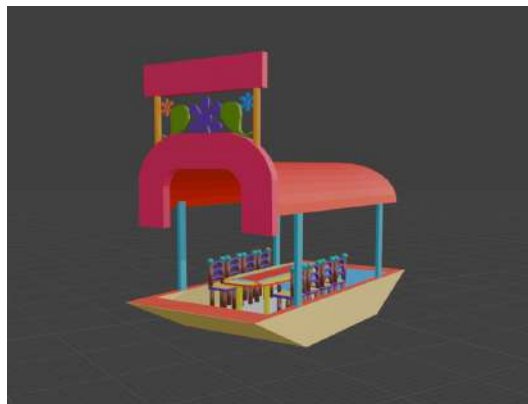


Image 37. Mini trajinera modeling.

5. Fan

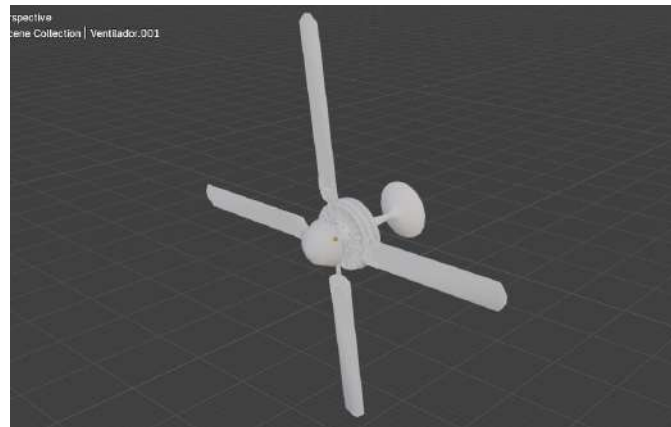


Image 38. Fan modeling.

6. Chart

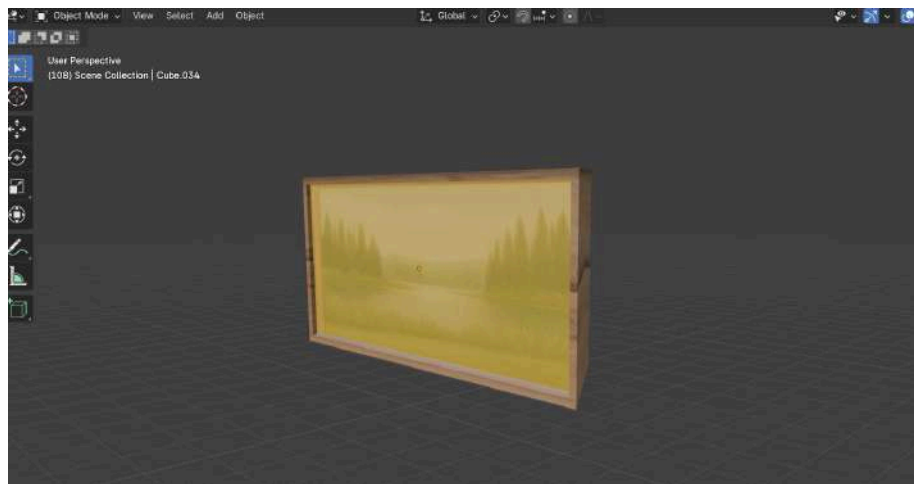


Image 39. Large box modeling.

7. Decorative paintings

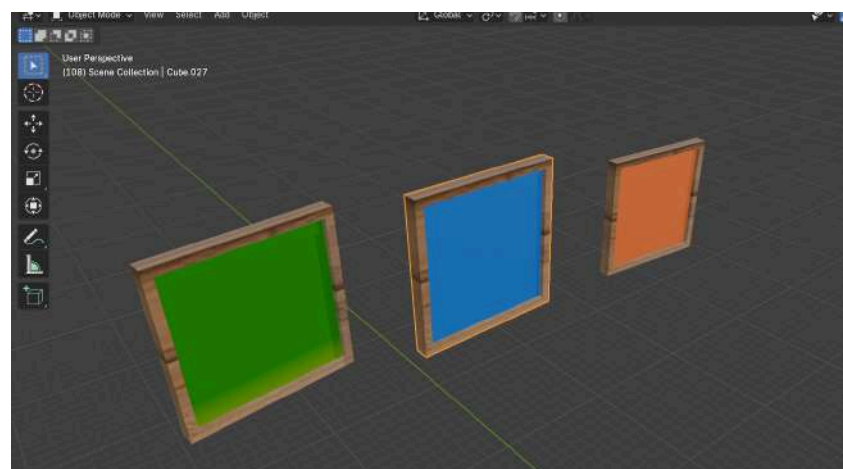


Image 40. Modeling decorative frames.

8. Door

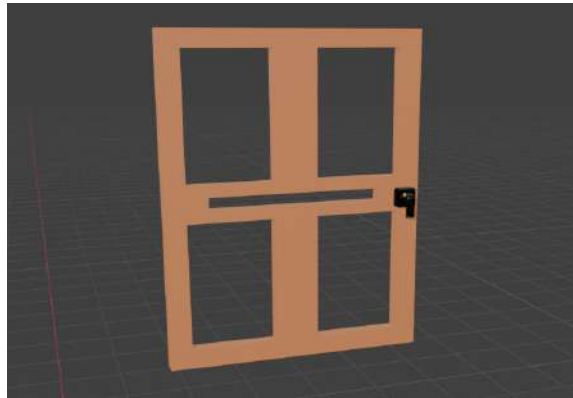


Image 41. Museum door modeling.

9. Stairs

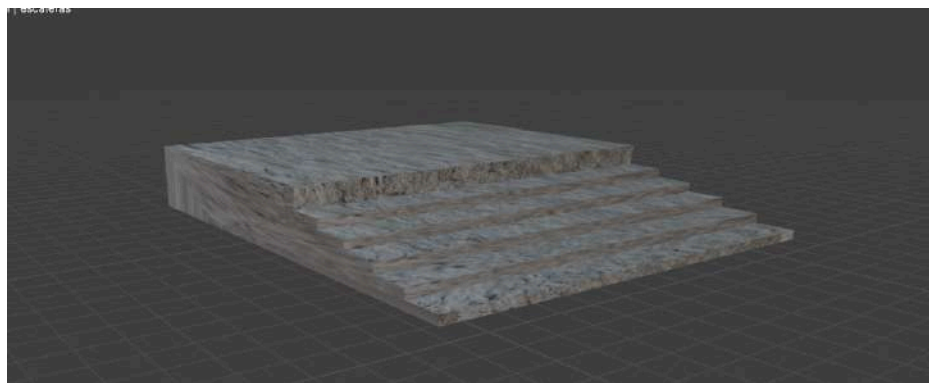


Image 42. Modeling of the Museum entrance stairs.

10. Bases for crafts

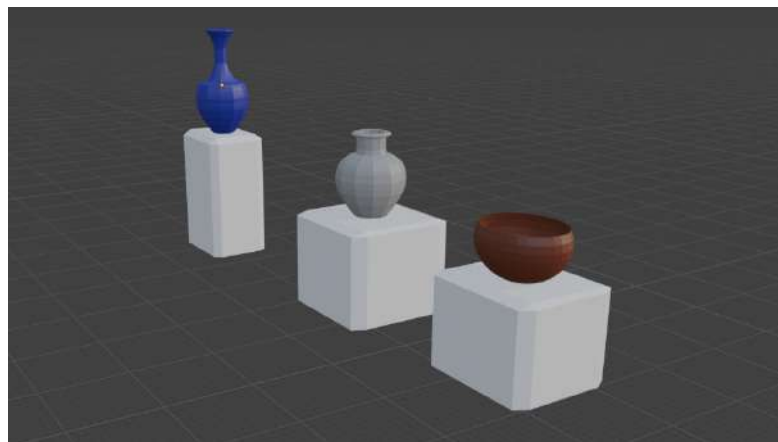


Image 43. Modeling bases for crafts.

11. Lamp



Image 44. Lamp modeling.

12. Mirror

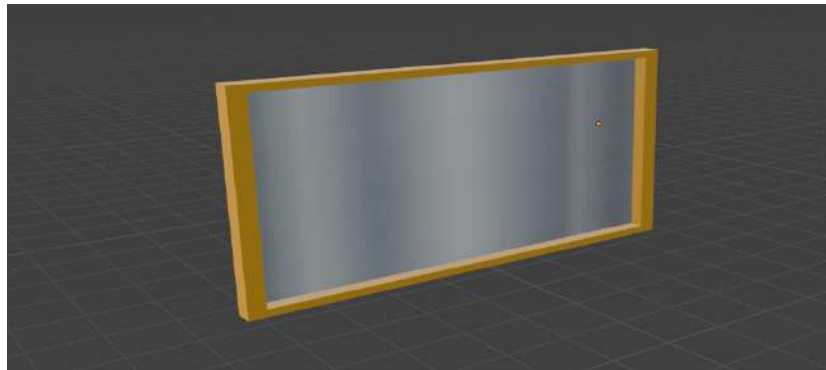


Image 45. Mirror modeling.

13. WC

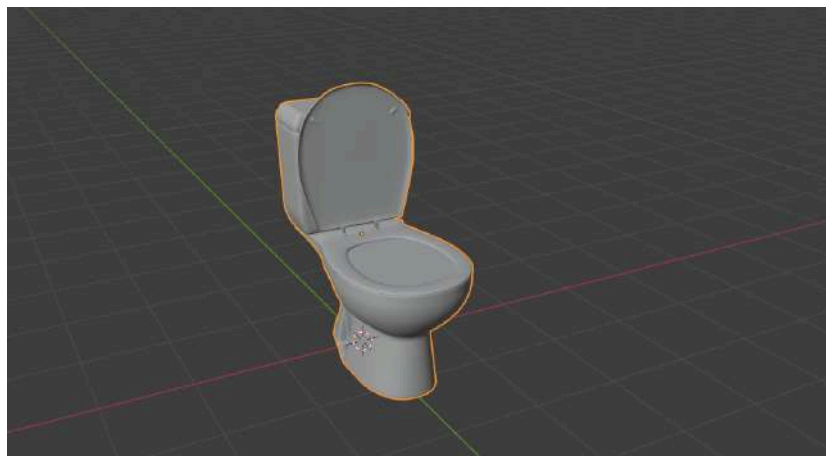


Image 46. WC modeling.

14. Bathroom doors

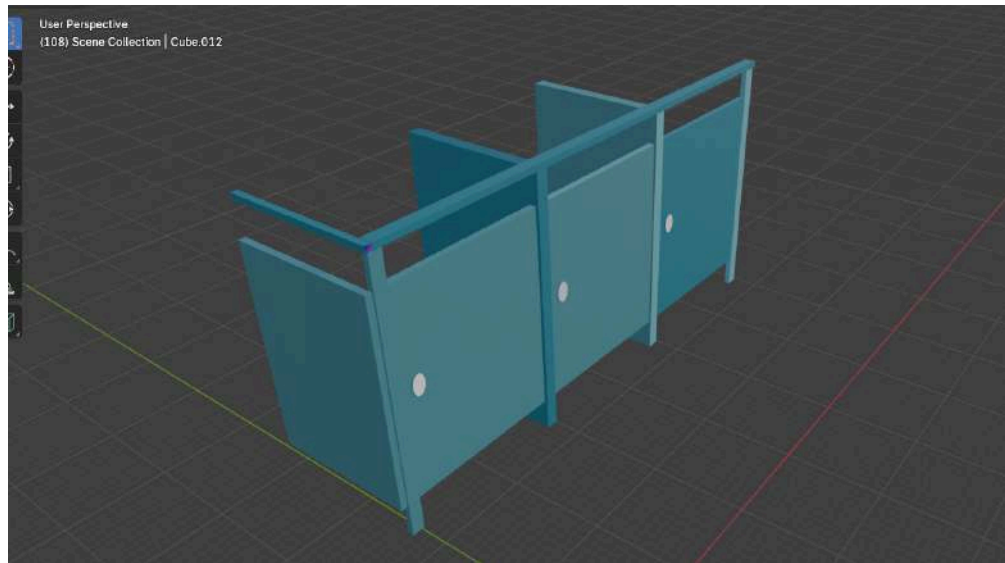


Image 47. Modeling bathroom doors.

15. Main bathroom door

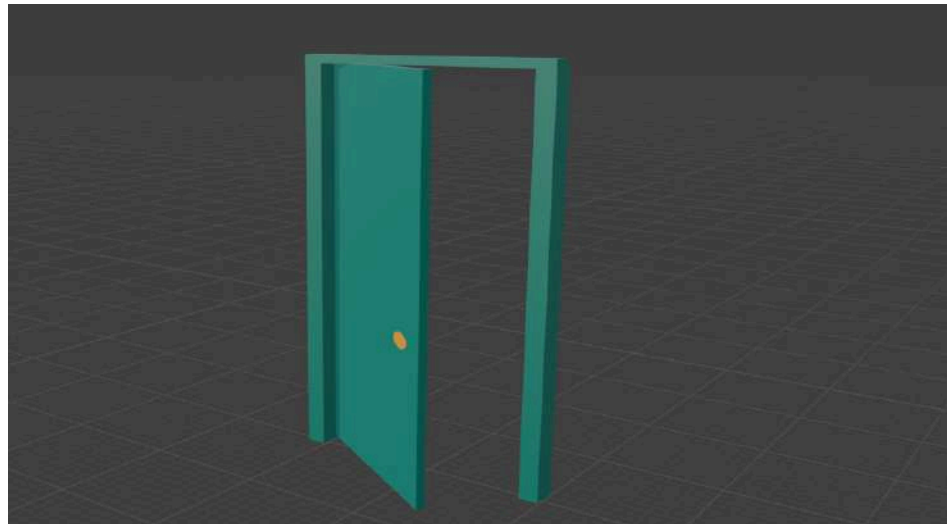


Image 48. Bathroom door modeling.

20. Sink

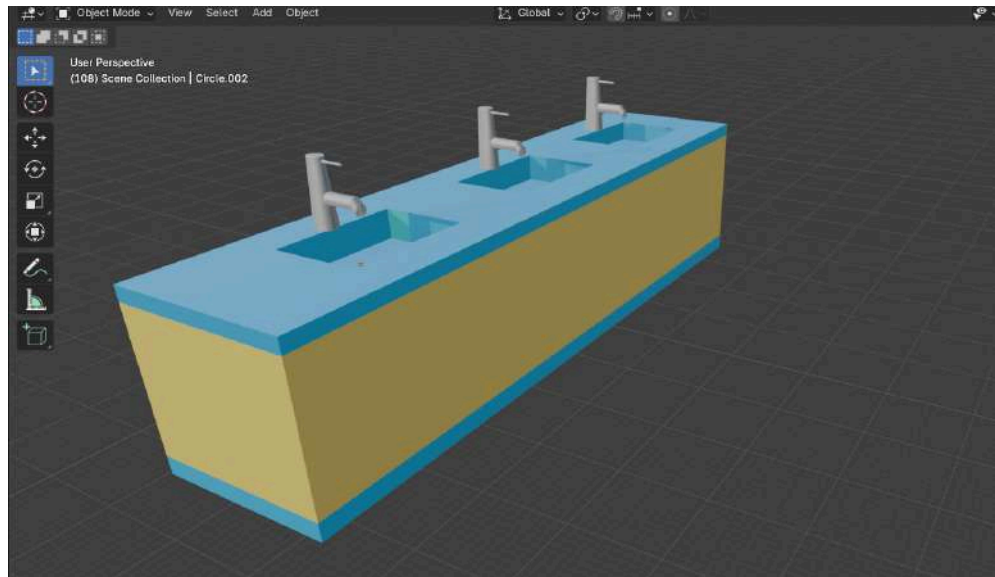


Image 49. Sink modeling.

21. Window

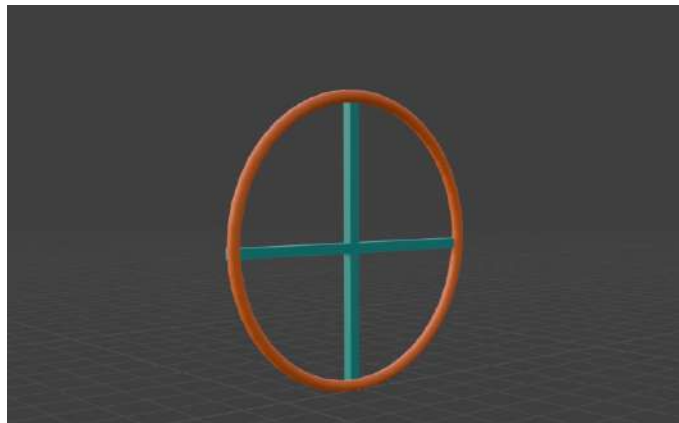


Image 50. Window modeling.

22. Ceiling objects

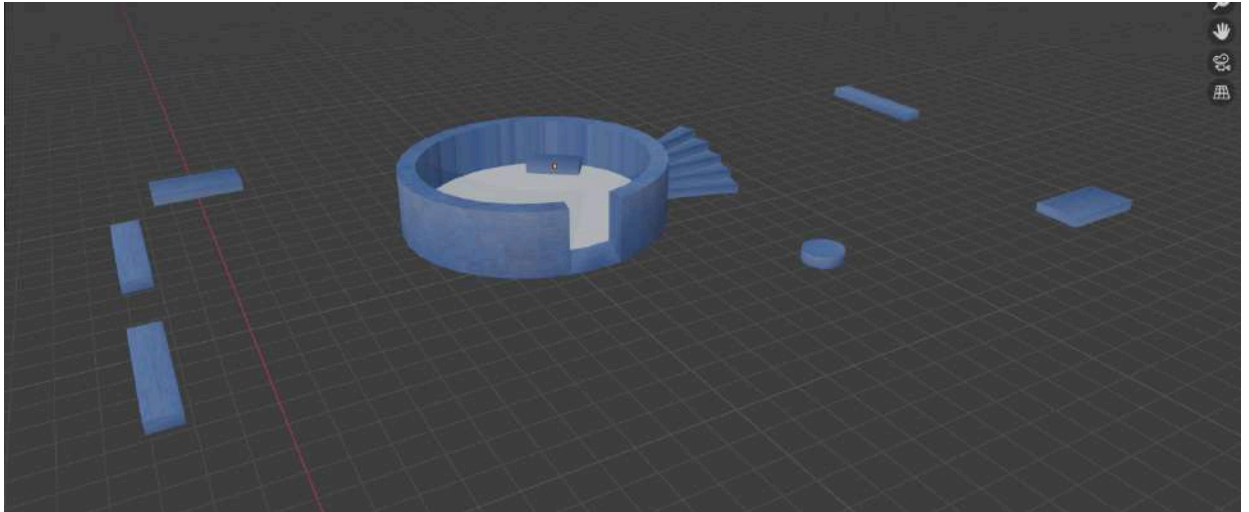


Image 51. Modeling parts of the Museum roof.

Park objects:

1. Ecological park sign



Image 52. Ecological park sign modeling.

2. Tower

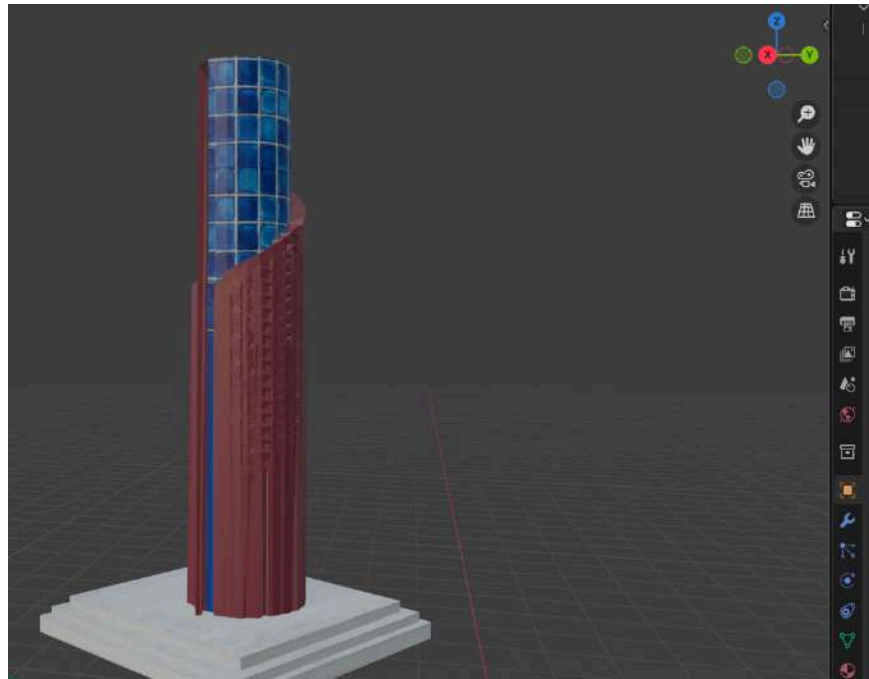


Image 53. Park tower modeling.

3. Bars

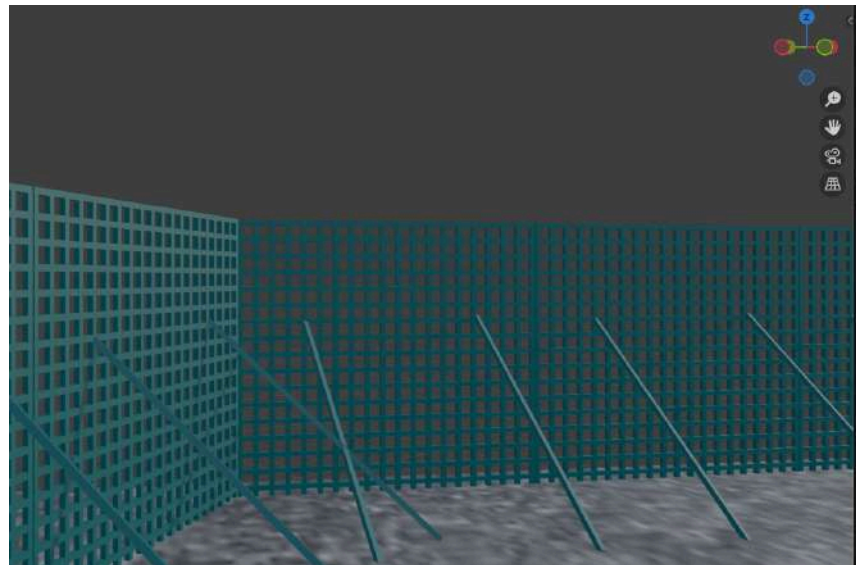


Image 54. Modeling of the fences that delimit the park.

4. Main entrance to the Park

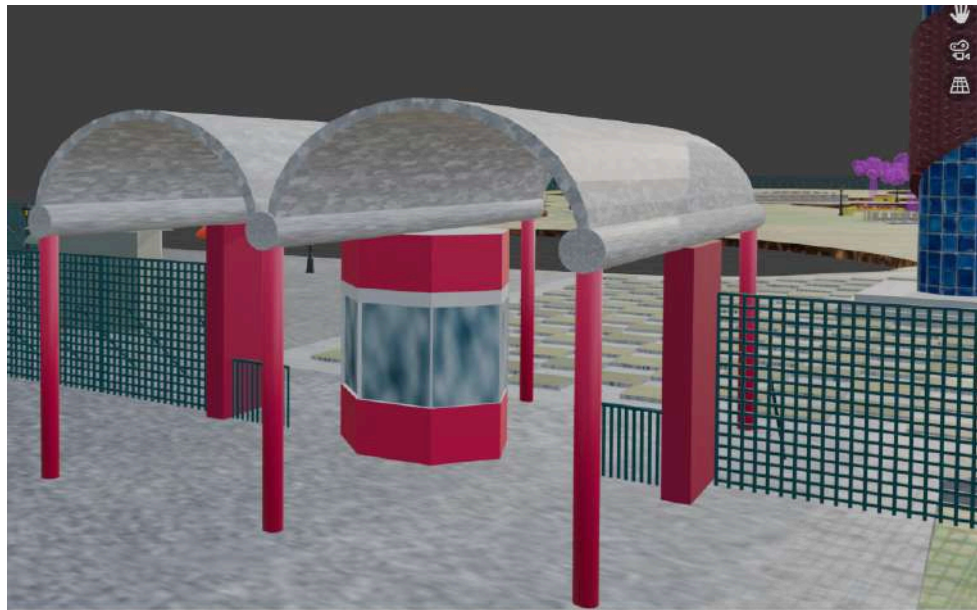


Image 55. Modeling of the park entrance.

5. Rest area

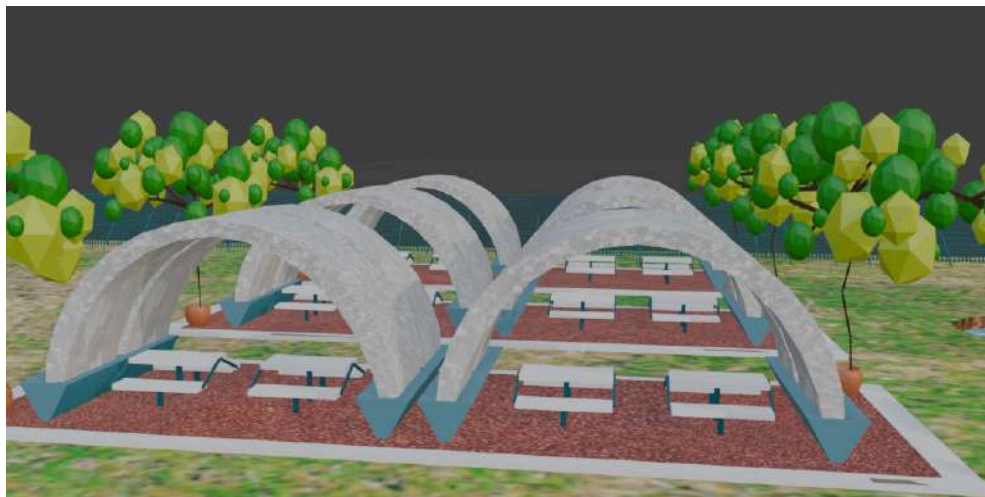


Image 56. Rest area model.

6. Bridge

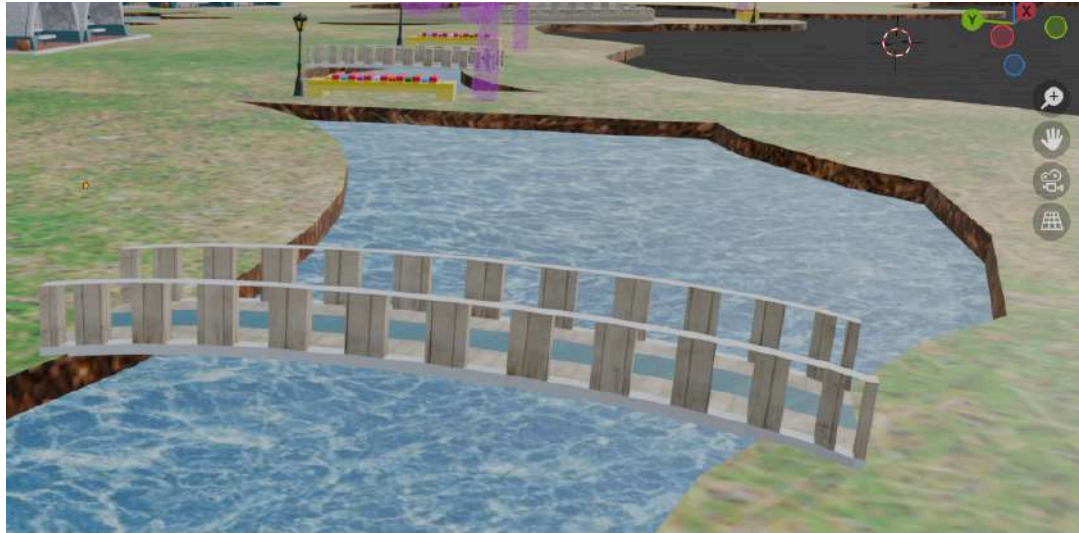


Image 57. Model referring to a bridge.

- List of models created by the team/taken from the internet:

Model	Authorship
Sauces	Model designed by team member.
Tepozanes	Model designed by team member.
Pumpkins	Model designed by team member.
Vegetable plantation	Model designed by team member.
Flower planting	Model designed by team member.
Poinsettia flowers	Model designed by team member.
Marigold Flowers	Model designed by team member.
Flores	Model designed by team member.
Trees	Model designed by team member.

Sparrows	Model designed by team member.
Eagles	Model designed by team member.
Exercise area	Model designed by team member.
Lagos	Model designed by team member.
Irregular settlements	Model designed by team member.
Dining room tables	Model designed by team member.
Food	Model designed by team member.
Museum memories	Model designed by team member.
Community market products	Model designed by team member.
Trajineras	Model designed by team member.
Decorations	Model designed by team member.
Ice cream cart	Model designed by team member.
Sales stands	Model designed by team member.
Field tables	Model designed by team member.
Stalls	Model designed by team member.
Reference books	Model designed by team member.
Museum documents	Model designed by team member.
Books for visitors	Model designed by team member.
Basketball	Model designed by team member.
Luminaires	Model designed by team member.
Chinampas	Model designed by team member.
Vases	Model designed by team member.
Bank	Model designed by team member.
Paintings	Model designed by team member.
Mini Trajinera	Model designed by team member.

Fan	Model designed by team member.
Chart	Model designed by team member.
Decorative Paintings	Model designed by team member.
Door	Model designed by team member.
Stairs	Model designed by team member.
Bases for crafts	Model designed by team member.
Lamp	Model designed by team member.
Mirror	Model designed by team member.
WC	Model designed by team member.
Bathroom doors	Model designed by team member.
Main bathroom door	Model designed by team member.
Washbasin	Model designed by team member.
Window	Model designed by team member.
Roof objects	Model designed by team member.
Ecological park sign	Model designed by team member.
Tower	Model designed by team member.
Bars	Model designed by team member.
Main entrance to the park	Model designed by team member.
Rest area	Model designed by team member.
Bridge	Model designed by team member.
Character	Model downloaded from the internet.

- Indicate the percentage of models created by the team vs. the percentage of models taken from the Internet:

The percentage of models made by the team was 96.77%.

The percentage of models taken from the Internet was 3.23%.

- integration of all models into one/multiple scenes (include titled screenshots of the integration from the modeling software, e.g., from Blender, Maya or 3DSMax as appropriate):

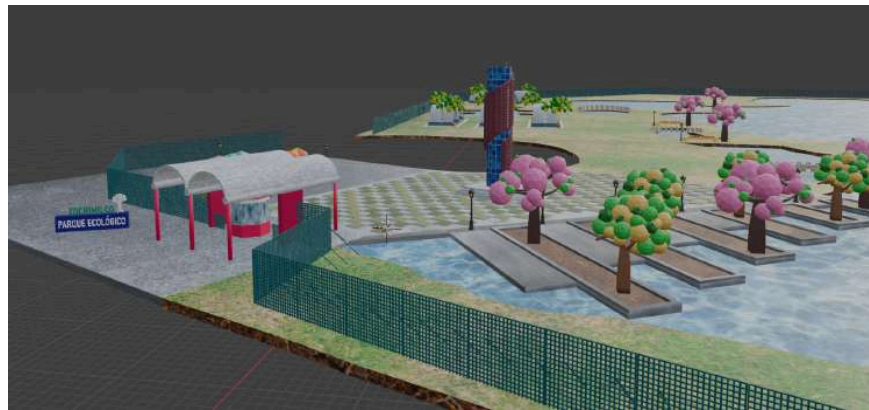


Image 58. Park entrance.



Image 59. View of the Lake and its surroundings.

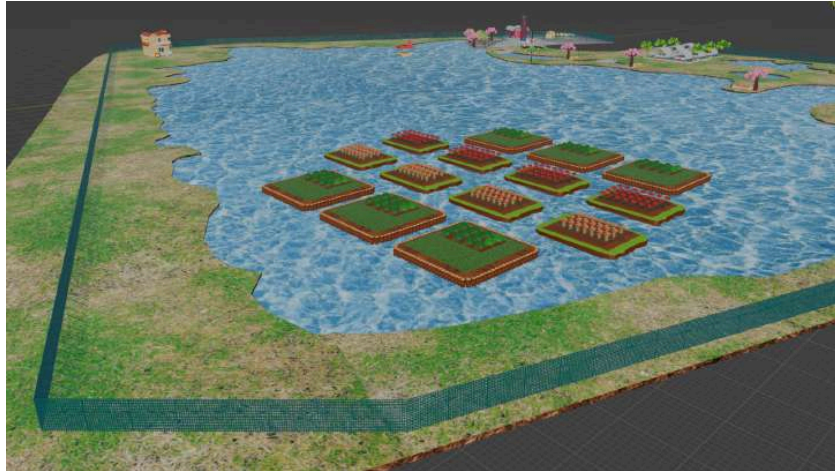


Image 60. Greenhouse.

Model classification

- List according to static and dynamic models:

Static models:

- Sauces
- Tepozanes
- Pumpkins
- Vegetable plantation
- Flower planting
- Poinsettia flowers
- Marigold flowers
- Flores

- Tree
- Sparrow
- Eagle
- Exercise area
- Lake
- Irregular settlements
- Dining room tables
- Food
- Museum memories
- Community market products
- Decorations
- Ice cream cart
- Sales stands
- Field tables
- Stalls
- Reference books
- Museum documents
- Books for visitors
- Basketball
- Luminaires
- Chinampas
- Bank
- Paintings
- Mini trajinera
- Chart
- Decorative paintings
- Stairs
- Bases for crafts
- Mirror
- WC
- Bathroom doors

- Main bathroom door
- Washbasin
- Window
- Ceiling objects
- Ecological park sign
- Tower
- Bars
- Main entrance to the park
- Rest area
- Bridge

Dynamic models:

- Trajinera
- Character
- Vase
- Museum Gate
- Fan
- Lamp

- List of models by type of material:

Marble:

- WC
- Washbasin
- Stairs
- Bases for crafts
- Main bathroom door
- Some parts of the museum

Wood:

- Sauces
- Tepozanes
- Trees
- Stalls
- Dining tables
- Field tables
- Sales stands
- Museum door
- Bathroom doors
- "Xochimilco" sign
- Pictures (frames)
- Reference books and visitors' books
- Decorations
- Bases for crafts

Weather:

- Decorative paintings
- Vases
- Museum memorabilia
- Lamp

Bronze:

- Luminaires
- Sculpture bases

Copper:

- Ice cream cart
- Ceiling objects
- Fan

Glass:

- Window
- Mirror
- Top of display cases if modeled
- Vases
- Paintings

- **List for dynamic elements: animation type (basic animation, procedural animation, keyframe-based animation, or other, for each dynamic 3D model.)**

1. Keyframe-based animation

We have our main character, who will have a keyframe animation, thanks to his armor.

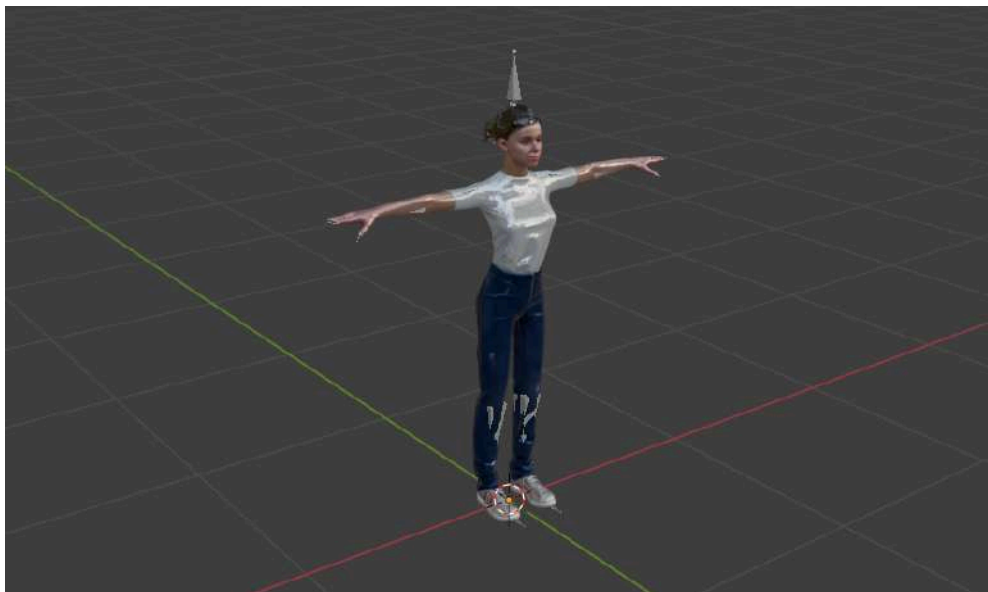


Image 61. Chosen character.

The objective of this animation is for our character to see the movement of the trajinera and, at the same time, follow it with his gaze and point at the trajinera.

In image 62 the character is in an initial posture with both arms relaxed and slightly bent, while maintaining a neutral standing position.



Image 62. Character armor.

The right hand is extended forward with the fingers individually rotated to create the pointing gesture.

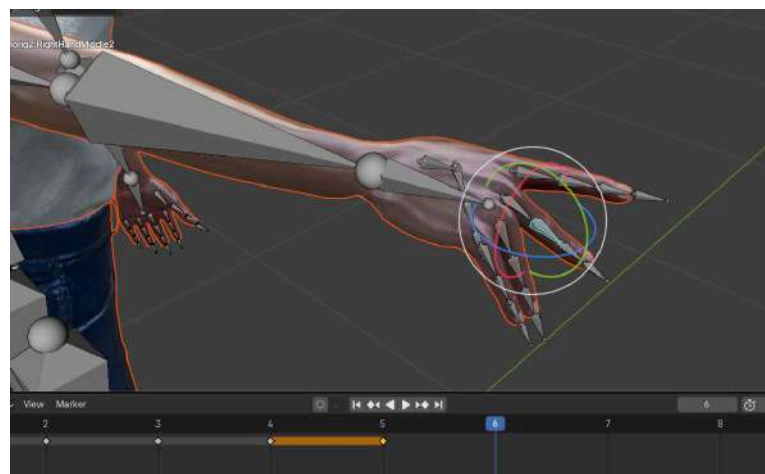


Image 63. Hand movement.

The character is shown from a rear view, holding his right arm extended in front of him and his index finger raised, completing the pointing gesture. This posture reinforces the character's gaze and communicative intent within the animation.

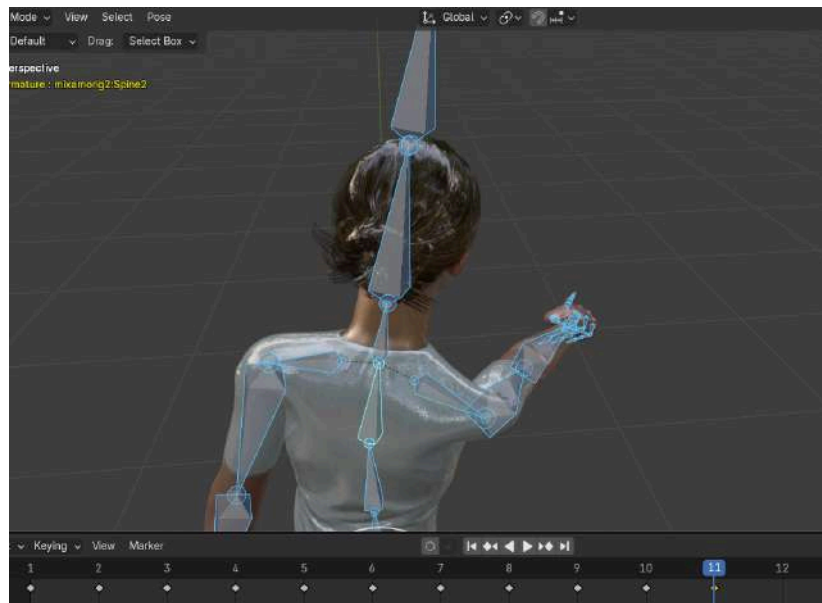


Image 64. Character animation.

Here our character is in the act of walking, with one leg raised forward and arms swinging naturally.



Image 65. Character animation.

In this final shot, the subject adopts a posture of active observation: his right arm is bent, his hand raised to his forehead, as if shielding himself from the sun or focusing his gaze into the distance. This pose clearly communicates attention or curiosity, reinforcing the idea that he is observing something passing by, such as a trajinera (a trajinera) from his position in the park.

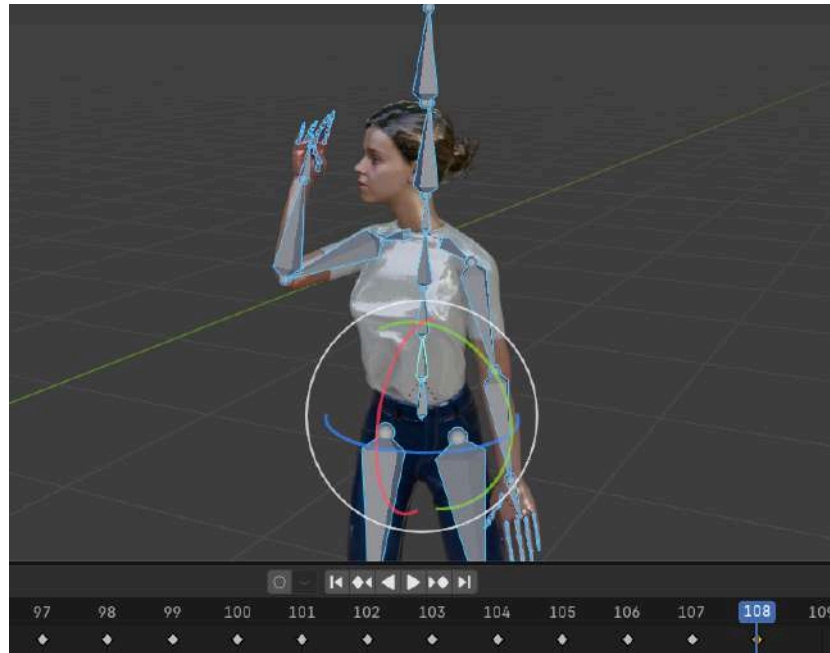


image 66. Final character animation.

Hierarchical Modeling

- List of all models that require a hierarchy to describe their structure. For example: characters, cars, mechanical systems, machinery, etc.
- Character
- Trajinera

- Include screenshots of the geometric model and its hierarchy. Use the bicycle or humanoid figure examples seen in class as a reference.

The hierarchical model of the trajinera is presented below, when it is in motion:

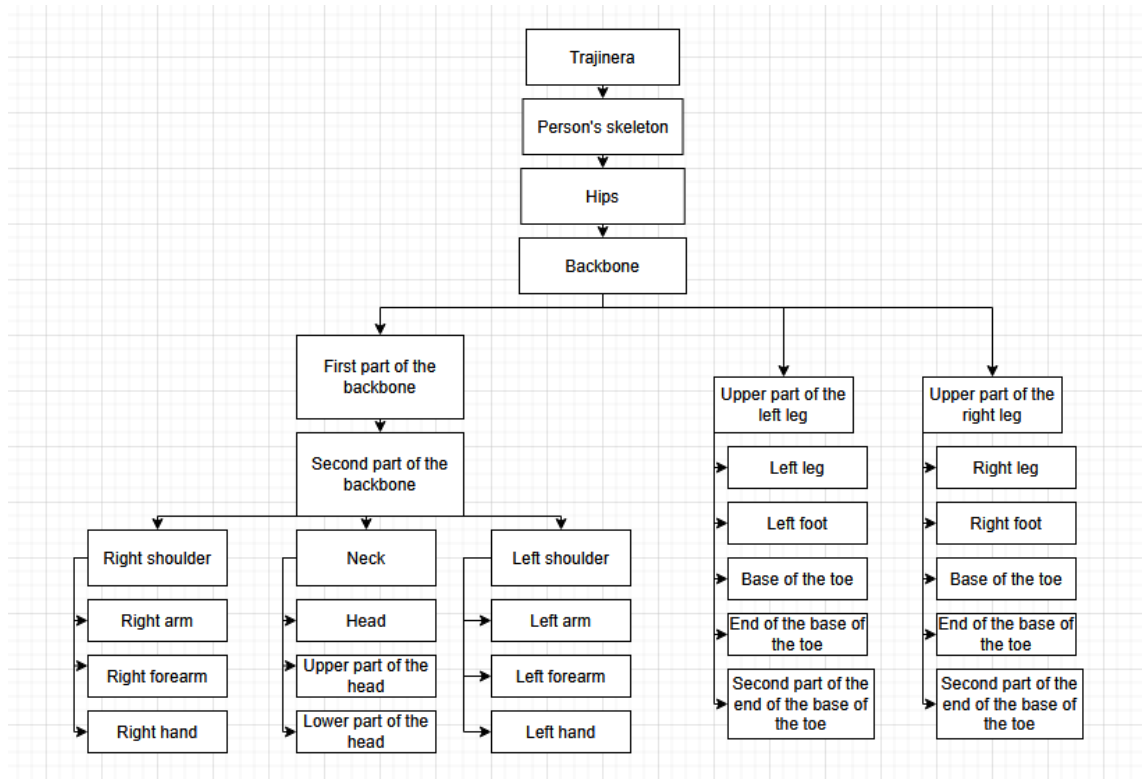


image 67. Hierarchical model of when the trajinera advances.

The hierarchical model allows you to animate a human skeleton composed of interconnected segments, where the movement of one part influences the others that depend on it. Branching out from the spine are limbs and joints such as arms, legs, head, and fingers.

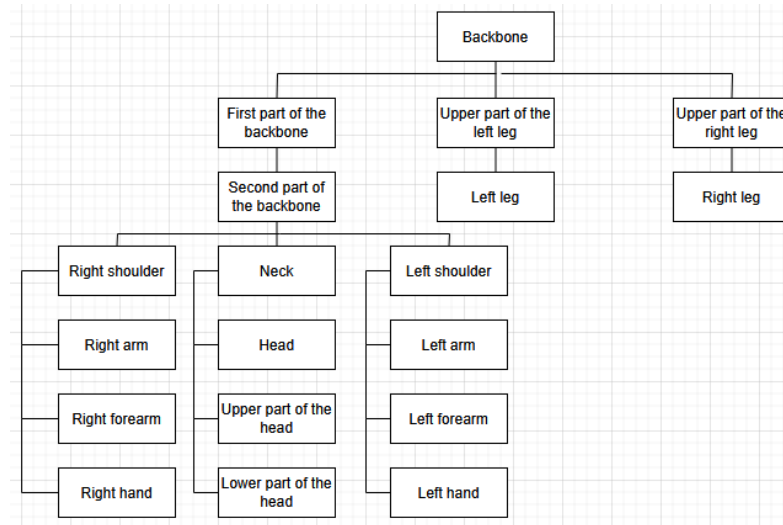


Image 68. Hierarchical model of the animation movement of when the character has movement.

This hierarchical model organizes the human body starting with the spine as the root node, from which structures such as arms, legs, and head branch off. Thanks to this hierarchy, complex movements such as raising an arm or turning the head are efficiently controlled, as each segment inherits the transformation from the previous one. This allows the animation to look fluid and consistent with the natural structure of the human body.

Artbook of 3D models with lighting effects

Material properties used by geometry:

Color:

- Each model was assigned a specific palette that respected the natural colors of the surroundings. For example, shades of green and ochre were used for the vegetation, and vibrant colors were used for the trajineras and decorations to maintain the cultural essence of Xochimilco.

Texture:

- Realistic textures created and edited in software such as GIMP and Krita were applied, including wood for benches and tables, marble for sanitary fixtures, glass for windows and vases, and metallic materials such as copper and bronze for decorative objects. The textures were UV mapped.

Visibility:

- All models were optimized to maintain clear visibility at different camera distances, with adjustments to culling and rendering order in OpenGL. Effects were used *depth testing* to avoid incorrect visual overlays.

Depth:

- The scene used a depth buffer to achieve spatially correct rendering. This allowed elements to be hidden or revealed based on their relative location, enhancing immersion.

To achieve these lighting characteristics, we use different components of the lighting sources such as specularity, ambient and diffuse lighting from the Phong lighting model to represent different materials that we want to emulate in the geometric models.

Prototypes of the models in OpenGL:

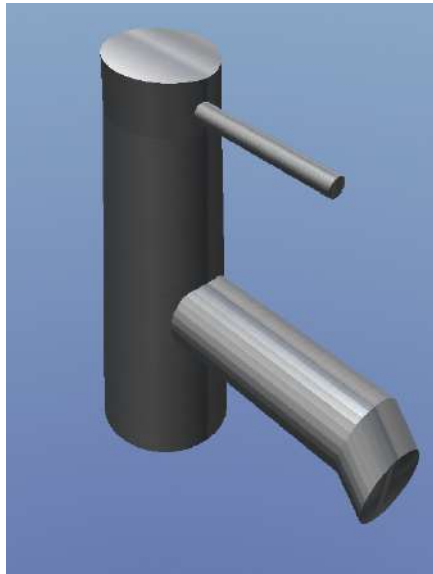


Image 69. Sink.



Image 70. Tree.



Image 71. Trajinera.

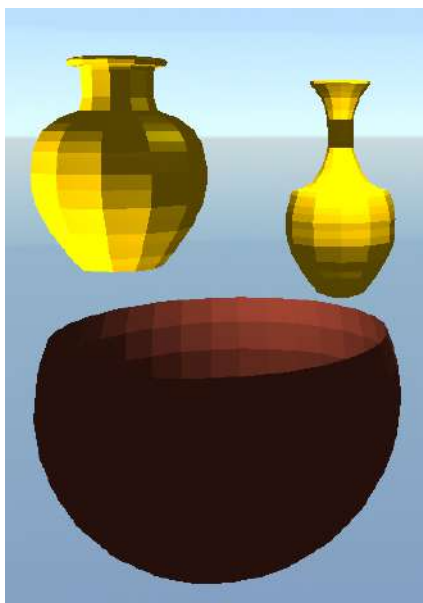


Image 72. Vases.



Image 73. Door.



Image 74. Lamp.

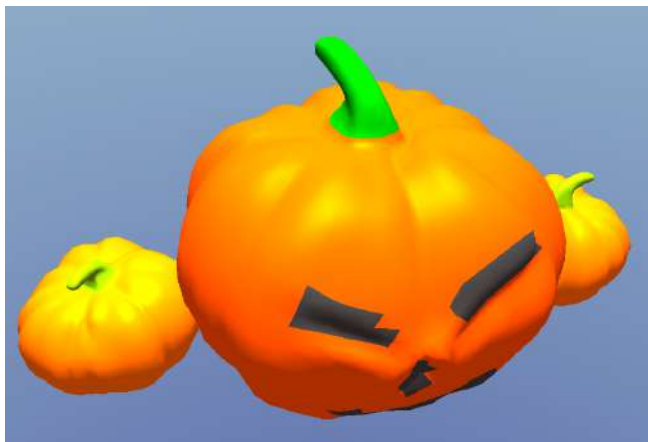


Image 75. Pumpkins.



Image 76. Banking.



Image 77. Water.

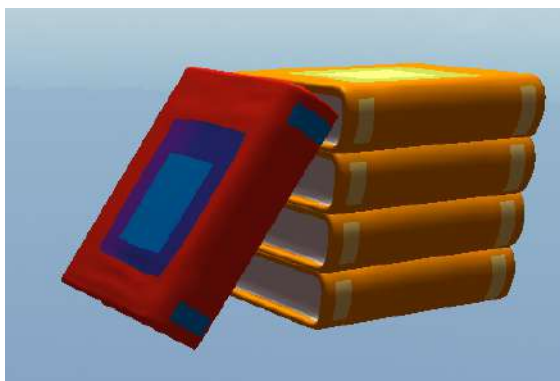


Image 78. Books.



Image 79. Tree.



Image 80. Basketball Basket.



Image 81. Market tent

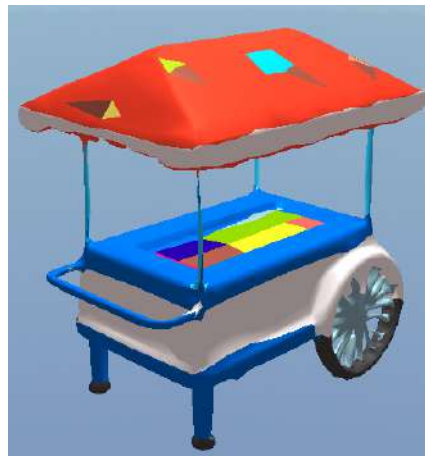


Image 82. Ice cream cart.



Image 83. House.

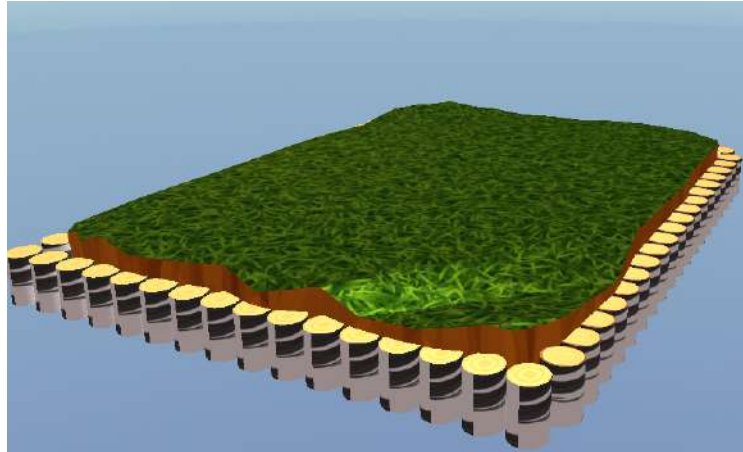


Image 84. Chinampa



Image 85. Swing.

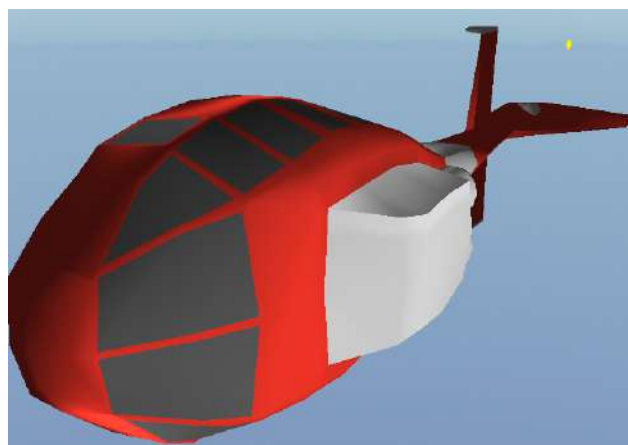


Image 86. Helicopter.

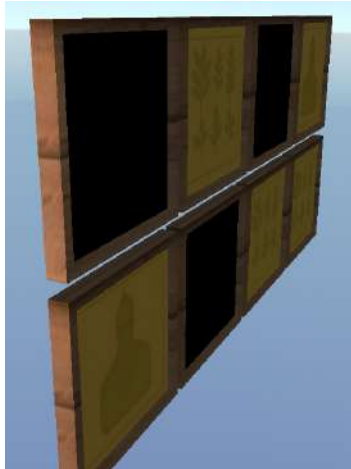


Image 87. Paintings.

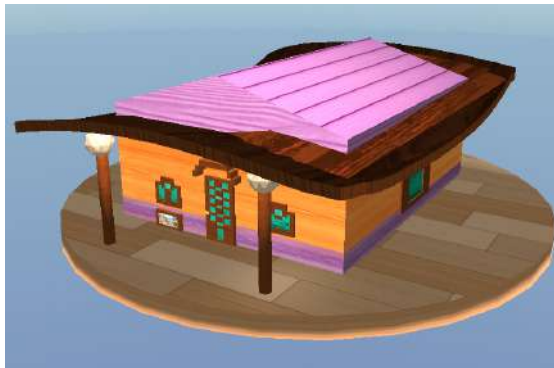


Image 88. School.



Image 89. Lantern.

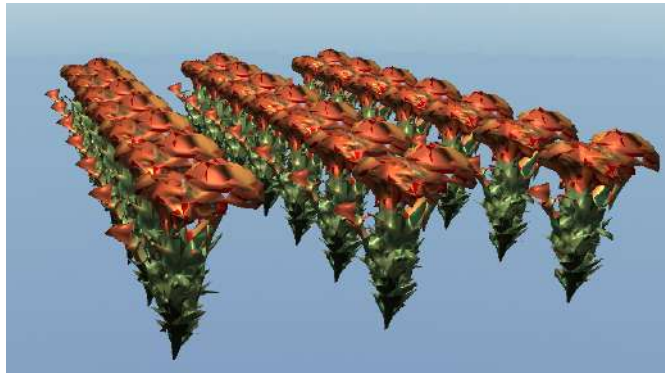


Image 90. Flowers.

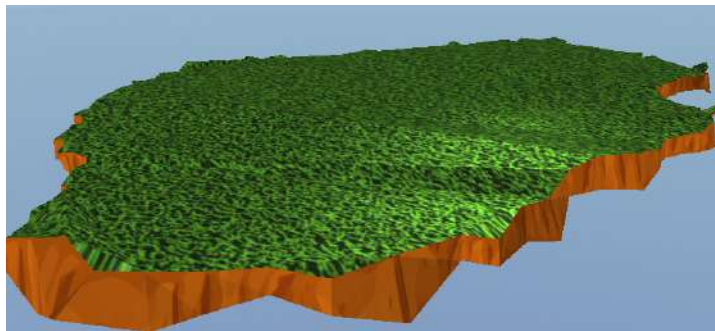


Image 91. Island.



Image 92. Table with umbrella.

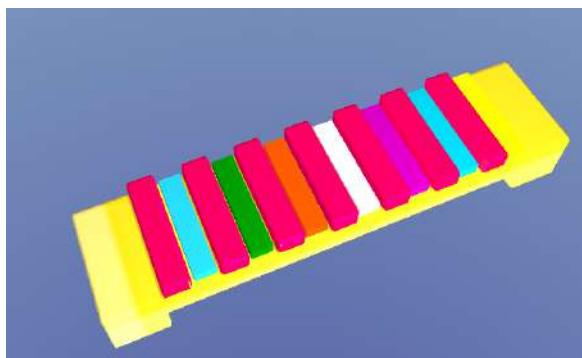


Image 93. Xylophone.

Table of Materials and Effects of the model

Model	Material Principal	Color Base	Applied Texture	Applied Lighting
Sauces	Wood.	Dark green	bark texture	diffuse sunlight
Trajinera	Painted wood.	Multicolor	hand painted	projected shadow + solar reflection
Vases	Crystal and Gold.	Gold and blue	textured enameled	interior spotlight
Museum Gate	Wood.	light brown	wood	ambient light
Lamp	Gold and Metal.	Dorado	shiny metal	oscillating light
Pumpkins	vegetable texture.	light green	smooth texture	ambient shadow
Stalls	Wood	wood	rough vegetation	Ambient lighting
Lake	Water with reflection	intense blue	moving water texture	reflection and transparency
xylophone	Plastic	Yellow	metal	lower

				brightness
The road	Plastic	White	Plastic	Dim glow
Washbasin	Metal	Silver	Metallic texture	Reflection
Market Tent	His	Red	Solid color	Spot light and shadow.
Ice cream cart	Painted metal.	White.	Solid color.	Brightness and shadow.
Home	Wood.	Light brown and yellow.	Wood.	Ambient light and plenty of it.
Chinampa	Grass and soil.	Green.	Meal.	Ambient and spot light.
Swing	Painted metal.	Yellow.	Solid color.	Ambient light and reflection.
Helicopter	Painted metal.	Red.	Solid color.	Ambient light and reflection.
Paintings	Wood, fabric.	Café.	Wood.	Ambient light.
School	Concrete and wood.	Orange.	Solid color and wood.	Ambient light and shadow.
Lighthouse.	Painted metal.	Negro	Solid color.	Ambient light.
Flores.	Organic material.	Rosa.	Color solid.	Ambient light and shadow.
The same	Meal.	Green	Meal.	Ambient and spot light.
Table with umbrella	Plastic.	Blue.	Solid color.	Ambient light and shadow.

Applied Animation Pipeline Artbook

- Describe the narrative of the idea through storyreel (drawings):

The narrative of "Conserving Xochimilco: An Interactive Journey" unfolds as an immersive tour through the Xochimilco Ecological Park. The experience begins with the user arriving at the park entrance, marked by a welcoming sign. Moving forward, they discover an environment that combines nature, history, and environmental issues. The user explores key areas such as Lake Huetzalin, the Flor de Chinampas Museum, and chinampas represented in different states: restored and deteriorated. Along the way, they are presented with scenes showing endemic species such as the axolotl, areas affected by pollution, and the interaction of characters involved in conservation activities, such as chinampa rehabilitation and canal cleaning. These vignettes, sketched in the storyreel, allow for visual planning of the development of key scenes.

Storyreel: Preserving Xochimilco: An Interactive Journey (narrative).

1) Welcome to the Park	2) Environmental Culture Center
The user begins their tour in front of the monumental sign reading "Xochimilco, Ecological Park." This point marks the beginning of the interactive tour with a brief message about the importance of preserving this natural space.	The scene shows the main building with its modern architecture. Here, visitors can enter an interactive room with information about the park's biodiversity and history.

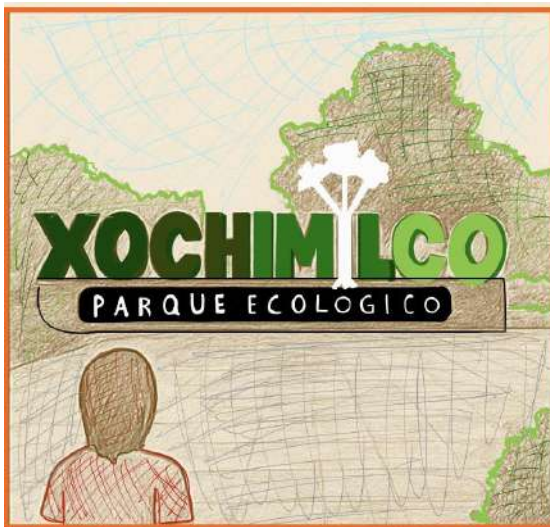


Image 94. Entrance to Xochimilco



Image 95. Cultural center of Xochimilco

3) Lake Acitlalin and its walkways

Visitors are shown walking along wooden walkways over the water, observing waterfowl and reed beds. The experience is immersive, with sounds of nature and animations of local fauna.

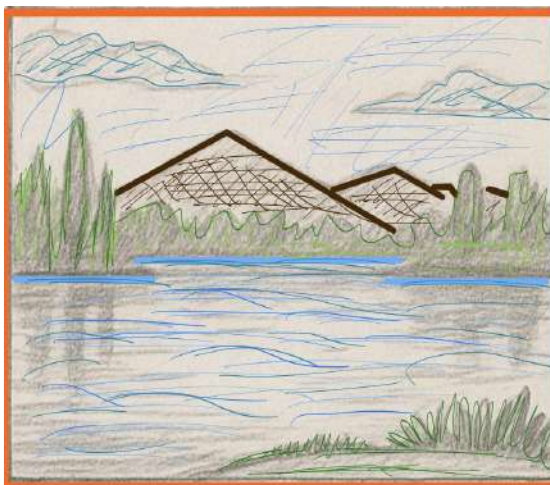


Image 96. Lake

4) Zone of chinampas and greenhouses

A scene where the user observes the restoration of chinampas and learns about traditional agricultural practices. Models of vegetables, cactus, and native flowers appear.



Image 97. View of the chinampas and greenhouses

5) Trajineras and Cuemanco pier

Visitors are shown near the pier with decorated trajineras. Here, they can embark on a virtual boat tour to explore the canal's flora, fauna, and environmental challenges.



Image 98. Trajineras

6) Flower walk and ahuejotes viewpoint

The tour concludes with a stroll between flower arches and tall trees. From an elevated viewing platform, visitors can admire the entire park, reinforcing the conservation message.



Image 99. Tours of the flower walk and viewpoint

7) Chinampas Flower Museum, Chinampaxochitl

It offers an immersive experience in the local flora and fauna, including ornamental plants, ahuejotes, pines, and various animals such as snakes, rabbits, and birds. The museum also allows you to learn about the history and traditions of the region.



Image 100. Entrance to the Chinampas Flower Museum

Storyreel: Dynamic model movement.



Puerta Cerrada

Image 101. Storyreel representation.



Puerta Semi-Abierta

Image 102. Storyreel representation.



Puerta Abierta

Image 103. Representation of storyreel movement.

- Applied animation process:

The animation process was carried out in different stages:

1. Planning and Storyreel:

- Key scenes were defined based on pedagogical and awareness-raising objectives.
- Drawings and sketches were made that guided the narrative structure of the environment.

2. Hierarchical modeling:

- Father-son relationships were established for models such as the trajinera (base, awning, oars) and the character (trunk, head, limbs).
- This allowed motion animations to be motion hierarchically and coherently.

3. Animation of dynamic models:

- Trajinera: linear interpolation animation for constant movement over the canal.
- Character: keyframe animation, where postures were defined at different moments of the action cycle (rowing, walking).

- Vases and fans: Basic rotation animations applied as environmental effects.
- Lamp: procedural animation with simulated oscillating motion.

4. Loading and rendering in OpenGL:

- The models were loaded using libraries such as Assimp.
- Geometric transformation was applied using matrices for translation, rotation and scaling.
- The animation system was integrated with the main rendering loop.

5. Shading and lighting:

- Basic lighting was applied by Lambert and Phong to highlight materials such as wood, marble and glass.
- Spot and directional lights were integrated to simulate the effect of the sun and park lights.

6. Synchronization with interactions:

- Events were programmed to trigger animations when the user approached certain areas, such as the start of movement of trajineras or the appearance of information panels about species.

This pipeline allows for a coherent and educational representation of the Xochimilco environment, encouraging interaction and active learning within an immersive virtual space.

FIRST PROTOTYPE

This section presents a preliminary prototype of our scene, which serves as a visual example of how the project will be structured. This prototype includes key elements of the environment, such as the representation of Xochimilco's characteristic flora, fauna, and infrastructure.

The goal is to provide a first look at the project's design and functionality, allowing for a feasibility assessment and adjustments before final development. This phase also helps identify improvements in interaction, graphic optimization, and educational elements that will contribute to raising awareness about ecosystem preservation.



Image 104. View of the “Flor de Chinampas” Museum and the Observation Tower.



Image 105. View behind the museum and the lake.

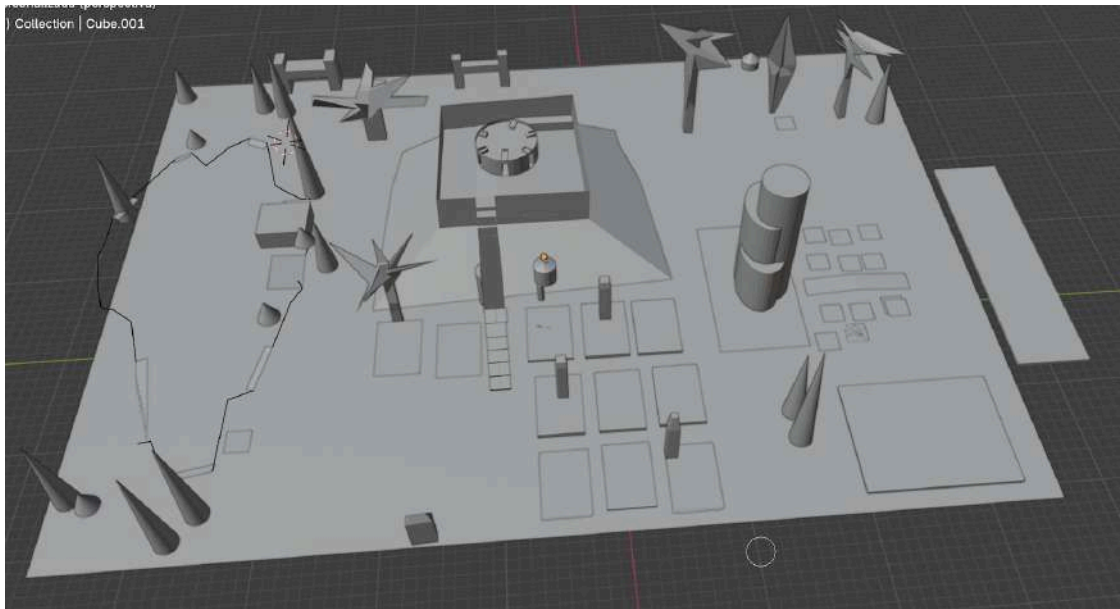


Image 106. View from Above.

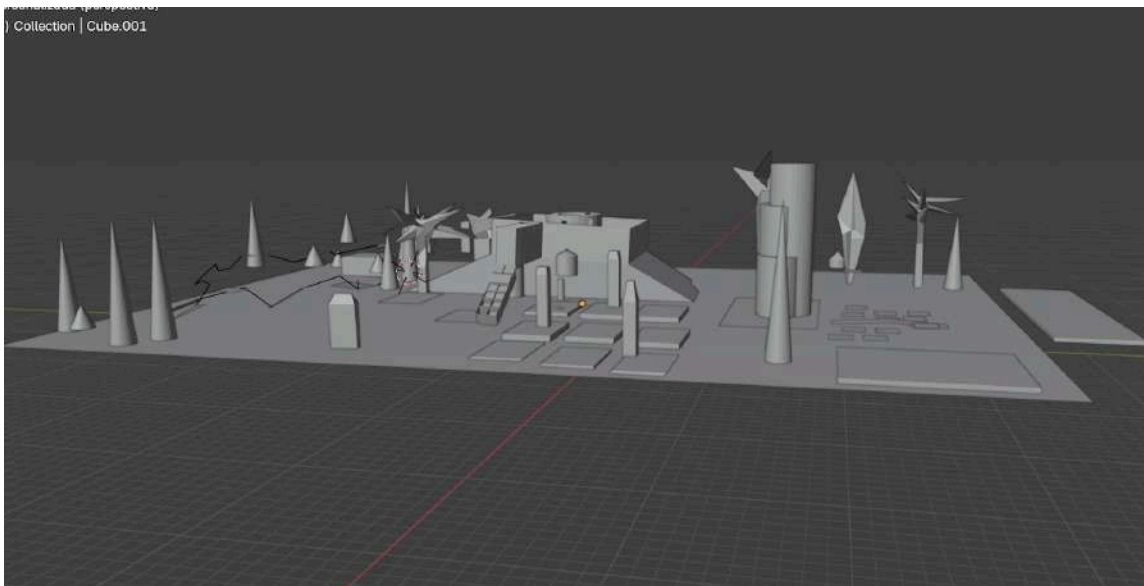


Image 107. Front View.

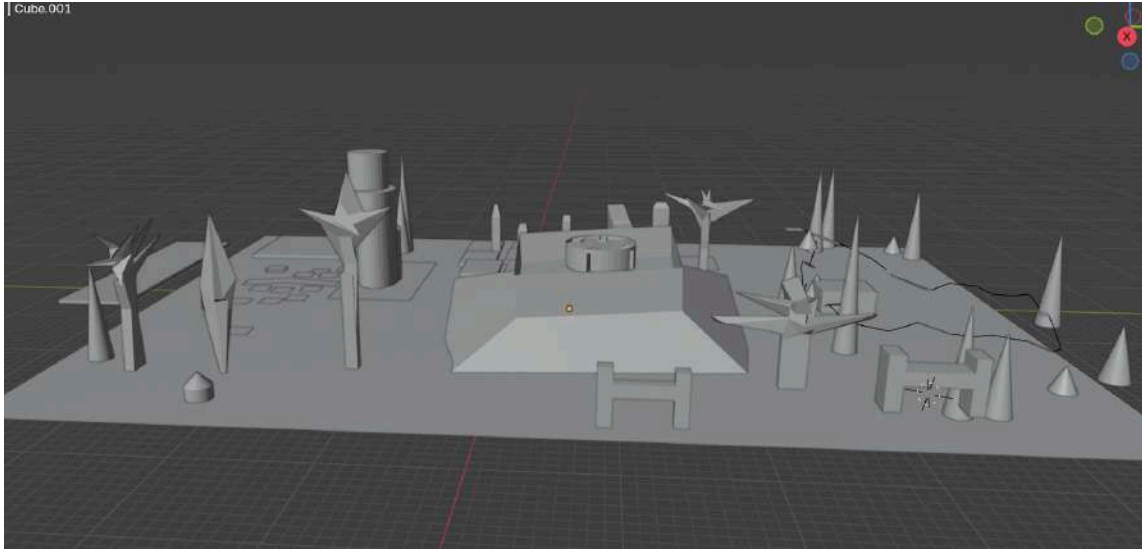


Image 108. Rear view and lake on the right

SECOND PROTOTYPE

Below is a prototype of the project in the design software used prior to loading models with OpenGL.

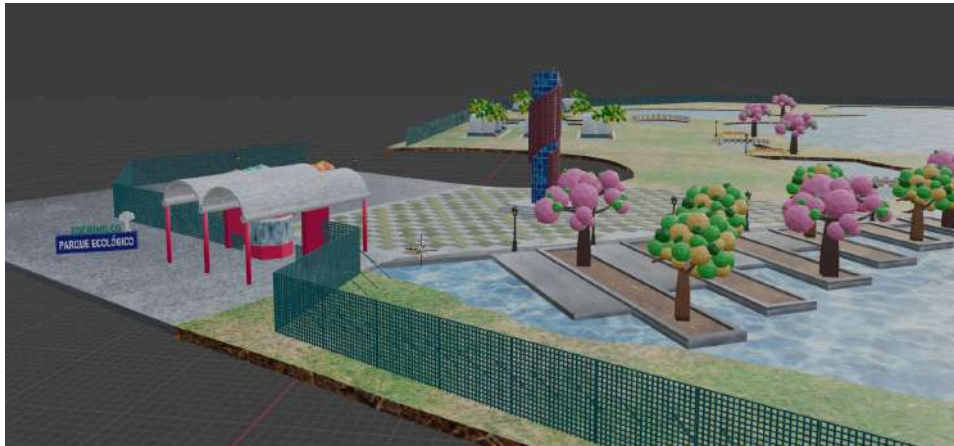


Image 109. Front view and lake on the right



Image 110. Left side view.

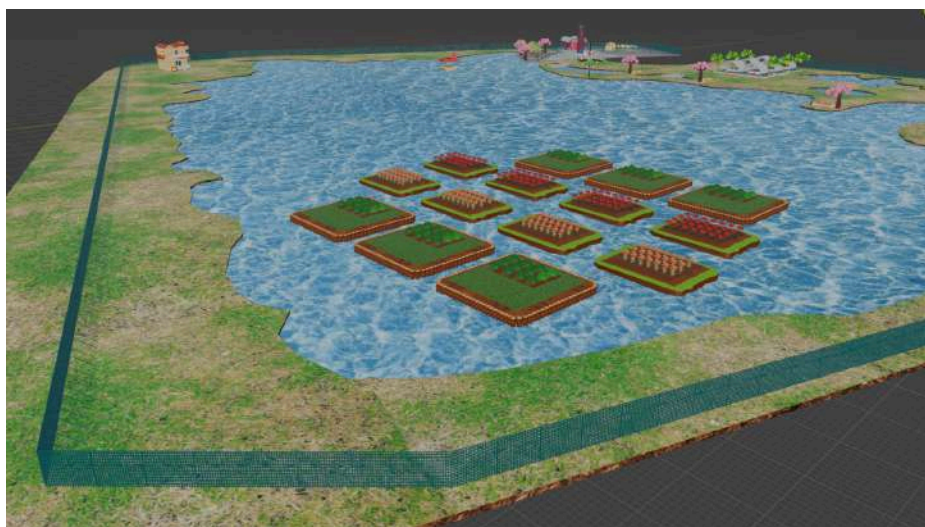


Image 111. Rear view and lake on the right.

We then load the models into the project.

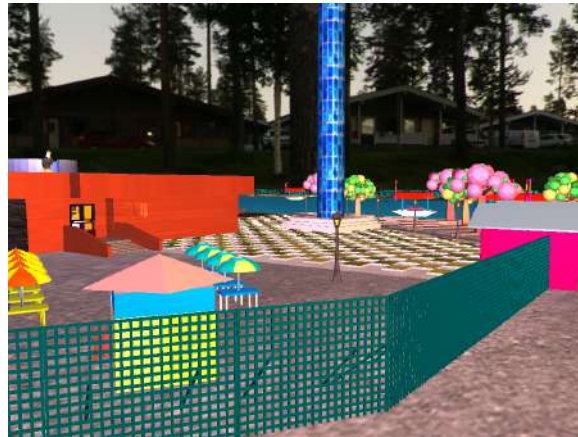


Image 112. Models loaded in OpenGL.

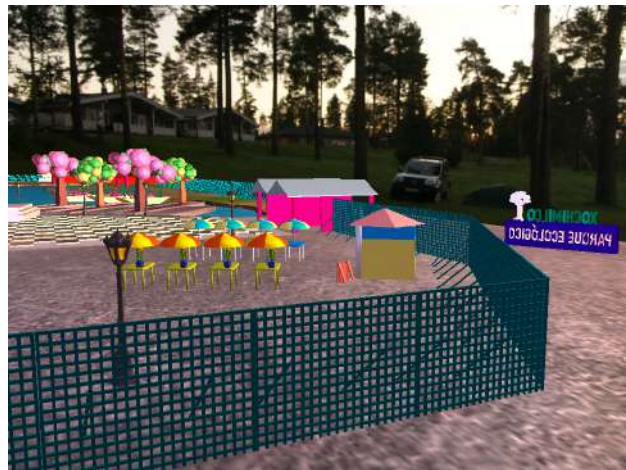


Image 113. Models loaded in OpenGL.



Image 114. Models loaded in OpenGL.



Image 115. Models loaded in OpenGL.



Image 116. Models loaded in OpenGL.

Software development

1.1. Software

For the implementation of the project, the following software tools were defined:

- Integrated Development Environment (IDE):
 - Visual Studio 2022 (Community or Professional version).

Justification:

- Native support for C/ languagesC++, necessary for the use of OpenGL and auxiliary libraries (Assimp for model loading).
 - Previous experience of the team in its management, acquired in the Computer Graphics and Human-Computer Interaction laboratory.
-
- Operating System:
 - Windows 10 or higher, as it is the target platform for development.
-
- Graphic Design Tools:
 - GIMP/Krita for texture design and user interface (UI) elements.
-
- Version Control:
 - GitHub for collaborative management of source code, 3D models, and supporting files.

1.2. Hardware

The following hardware resources are required:

- Development Teams:
 - 2 computers with: Windows 10/11.
 - 8 GB of RAM.
 - Dedicated graphics card (for GPU acceleration in the OpenGL rendering pipeline).

Purpose:

- Development and testing in different configurations (cross-testing).
- Optimizing heterogeneous computing for real-time geometry calculations.

1.3. Human Resources

The team is made up of 4 members, with the goal of achieving the following deliverables:

- Initial Planning and Documentation (stage completed).
- Active Development:
 - 3D modeling of missing objects.
 - Programming the graphic application (OpenGL/C++).
 - Flora/fauna research for accurate representation.
- Quality control:
 - Functionality and performance testing.
- Final Delivery:
 - Debugged executable.
 - Technical report and documentation.

1.4. Program structure.

The code follows a very similar structure to the programs we used in the lab class. The first section of code consists of several libraries that will facilitate the

construction of our project, including key elements such as the camera, functions for loading models, lighting, window creation, building the cubemap for the background, and other OpenGL dependencies.

```
#include <iostream>
#include <stdlib.h>

// GLAD: Multi-Language GL/GLSL/EGSL/GLX/WGL Loader-Generator
// https://glad.david.de/
#include <glad/glad.h>

// GLFW: https://www.glfw.org/
#include <GLFW/glfw3.h>

// GLM: OpenGL Math library
#include <glm/glm.hpp>
#include <glm/gtc/matrix_transform.hpp>
#include <glm/gtc/type_ptr.hpp>

// Model loading classes
#include <shader_m.h>
#include <camera.h>
#include <model.h>
#include <material.h>
#include <light.h>
#include <cubemap.h>
```

Image 117. Code structure.

Subsequently, prototypes of functions and variables are defined for the correct operation of the program, such as pointers to models, calculation of frame rate for animations, shaders, camera management and window size.

```
Model *arbol_1;
Model *arbol_2;
Model *arbol_3;
Model *banca_parque;
Model *luminaria;
Model *terreno;
Model *barda;
Model *letrero_parque;
Model *entrada_parque;
Model *torre;
Model *muelle;
Model *piso_entrada;
Model *puente_1;
Model *puente_2;
Model *domos;
Model *lago_principal;
Model *lagos_secundarios;
Model *trajinera_1;
Model *trajinera_2;
Model *chinampa_1;
Model *ballon;
```

Image 118. Variables and code operation.

The main function is executed to call the Start function and then Update, these two are the main functions of the program.

On the one hand, Start is responsible for initializing ourpuertoview, compile shaders, load the models into the scene (including the background cubemap) and initialize the camera.

```
bool Start() {
    // Inicialización de GLFW

    glfwInit();
    glfwWindowHint(GLFW_CONTEXT_VERSION_MAJOR, 3);
    glfwWindowHint(GLFW_CONTEXT_VERSION_MINOR, 3);
    glfwWindowHint(GLFW_OPENGL_PROFILE, GLFW_OPENGL_CORE_PROFILE);

    // Creación de la ventana con GLFW
    window = glfwCreateWindow(SCR_WIDTH, SCR_HEIGHT, "Conservando Xochimilco", NULL, NULL);
    if (window == NULL)
    {
        std::cout << "Failed to create GLFW window" << std::endl;
        glfwTerminate();
        return false;
    }
    glfwMakeContextCurrent(window);
    glfwSetFramebufferSizeCallback(window, framebuffer_size_callback);
    glfwSetCursorPosCallback(window, mouse_callback);
    glfwSetScrollCallback(window, scroll_callback);

    // glad: Cargar todos los apuntadores
    if (!gladLoadGLLoader((GLADloadproc)glfwGetProcAddress))
    {
        std::cout << "Failed to initialize GLAD" << std::endl;
        return false;
    }

    // Activación de buffer de profundidad
    glEnable(GL_DEPTH_TEST);

    // Compilación y enlace de shaders
    cubemapShader = new Shader("shaders/10_vertex_cubemap.vs", "shaders/10_fragment_cubemap.fs");
    mirrorShader = new Shader("shaders/14_mirror.vs", "shaders/14_mirror.fs");
    staticShader = new Shader("shaders/10_vertex_simple.vs", "shaders/10_fragment_simple.fs");
}
```

Image 119. Sample main function.

```

//Carga de modelos del parque
arbol_1 = new Model("models/IllumModels/modelos_proyecto/arbol_rosa.fbx");
arbol_2 = new Model("models/IllumModels/modelos_proyecto/arbol_xochi_1.fbx");
arbol_3 = new Model("models/IllumModels/modelos_proyecto/arbol_xochi_2");
banca_parque = new Model("models/IllumModels/modelos_proyecto/");
terreno = new Model("models/IllumModels/modelos_proyecto/parque(terreno).obj");
luminaria = new Model("models/IllumModels/modelos_proyecto/");
barda = new Model("models/IllumModels/modelos_proyecto/");
letrero_parque = new Model("models/IllumModels/modelos_proyecto/");
entrada_parque = new Model("models/IllumModels/modelos_proyecto/");
torre = new Model("models/IllumModels/modelos_proyecto/");
muelle = new Model("models/IllumModels/modelos_proyecto/");
piso_entrada = new Model("models/IllumModels/modelos_proyecto/");
puente_1 = new Model("models/IllumModels/modelos_proyecto/");
puente_2 = new Model("models/IllumModels/modelos_proyecto/");
domos = new Model("models/IllumModels/modelos_proyecto/");
lago_principal = new Model("models/IllumModels/modelos_proyecto/");
lagos_secundarios = new Model("models/IllumModels/modelos_proyecto/");
trajinera_1 = new Model("models/IllumModels/modelos_proyecto/");
trajinera_2 = new Model("models/IllumModels/modelos_proyecto/");
chinampa_1 = new Model("models/IllumModels/modelos_proyecto/chinampas_flores.obj");
ballon = new Model("models/IllumModels/modelos_proyecto/");

// Cubemap
vector<std::string> faces
{
    "textures/cubemap/01/posx.png",
    "textures/cubemap/01/negx.png",
    "textures/cubemap/01/posy.png",
    "textures/cubemap/01/negy.png",
    "textures/cubemap/01/posz.png",
    "textures/cubemap/01/negz.png"
};
mainCubeMap = new CubeMap();
mainCubeMap->loadCubemap(faces);

camera3rd.Position = position;
camera3rd.Position.y += 1.7f;
camera3rd.Position -= forwardView;

```

Image 120. Loading park models.

After this, the Update function comes into play, which will be responsible for processing all the frames following the initial one, updating values such as the camera position, the position of the models, real-time transformations to objects, and drawing them according to how the user interacts with the program, modifying the parameters of the projection, translation, rotation, and scale matrices.

```

// Dibujo de la zona de chinampas
{
    staticShader->use();

    glEnable(GL_BLEND);
    glBlendFunc(GL_SRC_ALPHA, GL_ONE_MINUS_SRC_ALPHA);

    staticShader->setMat4("projection", projection);
    staticShader->setMat4("view", view);

    glm::mat4 model = glm::mat4(1.0f);
    model = glm::translate(model, glm::vec3(0.0f, -15.0f, 0.0f)); // translate it down so it's at the center of the scene
    model = glm::rotate(model, glm::radians(0.0f), glm::vec3(1.0f, 0.0f, 0.0f));
    model = glm::scale(model, glm::vec3(0.7f, 0.7f, 0.7f)); // it's a bit too big for our scene, so scale it down
    staticShader->setMat4("model", model);

    chinampa_1->Draw(*staticShader);

    model = glm::translate(model, glm::vec3(0.0f, 0.0f, 30.0f)); // translate it down so it's at the center of the scene
    model = glm::rotate(model, glm::radians(0.0f), glm::vec3(1.0f, 0.0f, 0.0f));
    model = glm::scale(model, glm::vec3(1.0f, 1.0f, 1.0f)); // it's a bit too big for our scene, so scale it down
    staticShader->setMat4("model", model);

    chinampa_1->Draw(*staticShader);
}

```

Image 121. Drawing chinampa area.

At the end of the file, we find the function to process the inputs that the user enters through the keyboard and mouse peripherals where we mainly manage various properties of the camera such as its position and rotation (where it points).

```

// Procesamos entradas del teclado
void processInput(GLFWwindow* window)
{
    if (glfwGetKey(window, GLFW_KEY_ESCAPE) == GLFW_PRESS)
        glfwSetWindowShouldClose(window, true);

    if (glfwGetKey(window, GLFW_KEY_W) == GLFW_PRESS)
        camera.ProcessKeyboard(FORWARD, deltaTime);
    if (glfwGetKey(window, GLFW_KEY_S) == GLFW_PRESS)
        camera.ProcessKeyboard(BACKWARD, deltaTime);
    if (glfwGetKey(window, GLFW_KEY_A) == GLFW_PRESS)
        camera.ProcessKeyboard(LEFT, deltaTime);
    if (glfwGetKey(window, GLFW_KEY_D) == GLFW_PRESS)
        camera.ProcessKeyboard(RIGHT, deltaTime);
    if (glfwGetKey(window, GLFW_KEY_M) == GLFW_PRESS)
        glPolygonMode(GL_FRONT_AND_BACK, GL_LINE);
    if (glfwGetKey(window, GLFW_KEY_N) == GLFW_PRESS)
        glPolygonMode(GL_FRONT_AND_BACK, GL_FILL);
    if (glfwGetKey(window, GLFW_KEY_B) == GLFW_PRESS)
        glPolygonMode(GL_FRONT_AND_BACK, GL_POINT);

    // camera movement
    if (glfwGetKey(window, GLFW_KEY_UP) == GLFW_PRESS) {
        position = position + 0.001f * forwardView;
        camera3rd.Front = forwardView;
        camera3rd.ProcessKeyboard(FORWARD, deltaTime);
        camera3rd.Position = position;
        camera3rd.Position.y += 1.7f;
        camera3rd.Position -= forwardView;
    }
    if (glfwGetKey(window, GLFW_KEY_DOWN) == GLFW_PRESS) {
        position = position - 0.001f * forwardView;
        camera3rd.Front = forwardView;
        camera3rd.ProcessKeyboard(BACKWARD, deltaTime);
        camera3rd.Position = position;
        camera3rd.Position.y += 1.7f;
        camera3rd.Position -= forwardView;
    }
}

```

Image 122. Code referring to camera movements.

Below is a simple flowchart illustrating the general operation of our code.

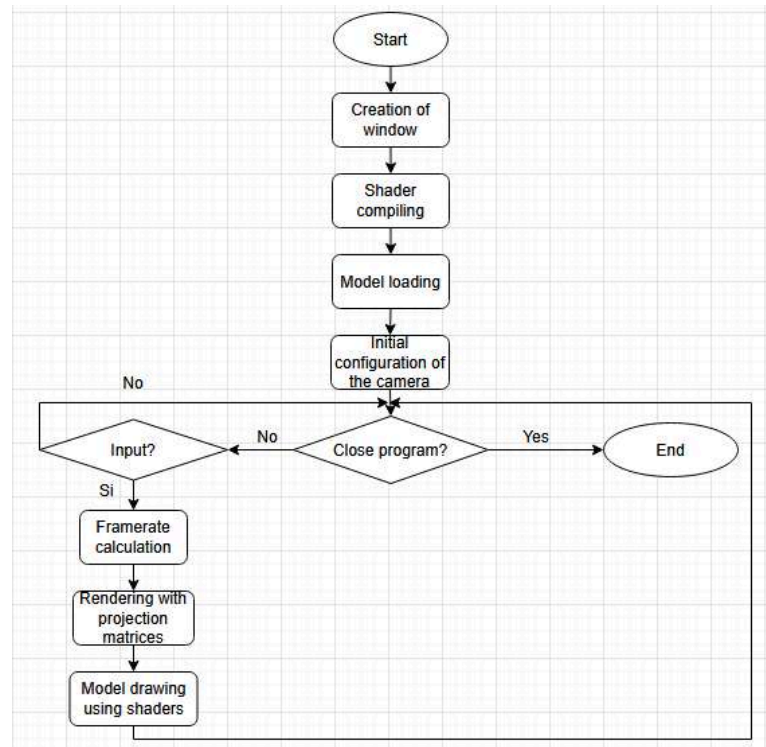


Image 123. Flowchart of the code operation.

EXPERIMENTS

1. Loading of land and chinampas

We first experimented by loading the terrain model and the chinampas to verify the operation of the model loading library.assimp and rule out any errors that may arise in this regard.

We first loaded the park land and a couple of chinampas, as we considered these models to be the fundamental basis of our project.

```
terreno = new Model("models/IllumModels/modelos_proyecto/parque(terreno).obj");  
  
chinampa_1 = new Model("models/IllumModels/modelos_proyecto/chinampas_flores.obj");
```

Image 124. Code for the loading of land and chinampas.

2. Loading animations.

Then we ran a test to verify how the animations worked in our project. To do this experiment, we used a simple animation that would be essential to bring our scene to life: the movement of the lake water.

This animation is included in the model so no code had to be modified and simply the program loaded.

```
lago_principal = new Model("models/IllumModels/modelos_proyecto/");
```

Image 125. Code for loading animations.

3. View in mesh mode and point mode.

With the models loaded into our scene, we can see their structure in the form of meshes or points. This experiment is quite useful for visualizing the internal structures of several of our models and determining if we have too many polygons that could affect the operation of our program to decide if it would be convenient to reduce the geometry count to optimize the models.

```
if (glfwGetKey(window, GLFW_KEY_M) == GLFW_PRESS)
    glPolygonMode(GL_FRONT_AND_BACK, GL_LINE);
if (glfwGetKey(window, GLFW_KEY_N) == GLFW_PRESS)
    glPolygonMode(GL_FRONT_AND_BACK, GL_FILL);
if (glfwGetKey(window, GLFW_KEY_B) == GLFW_PRESS)
    glPolygonMode(GL_FRONT_AND_BACK, GL_POINT);
```

Image 126. Code for loading the scene.

RESULTS

Modeling:

The park's representative elements were modeled using Blender, using geometric primitives and subdivision, extrusion, and digital sculpting techniques. Featured objects include the following:



Image 127. Sign modeling.

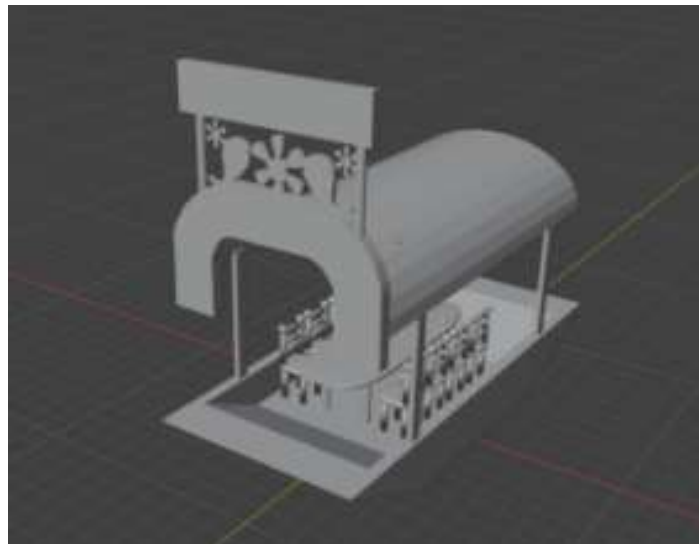


Image 128. Modeling of a trajinera.

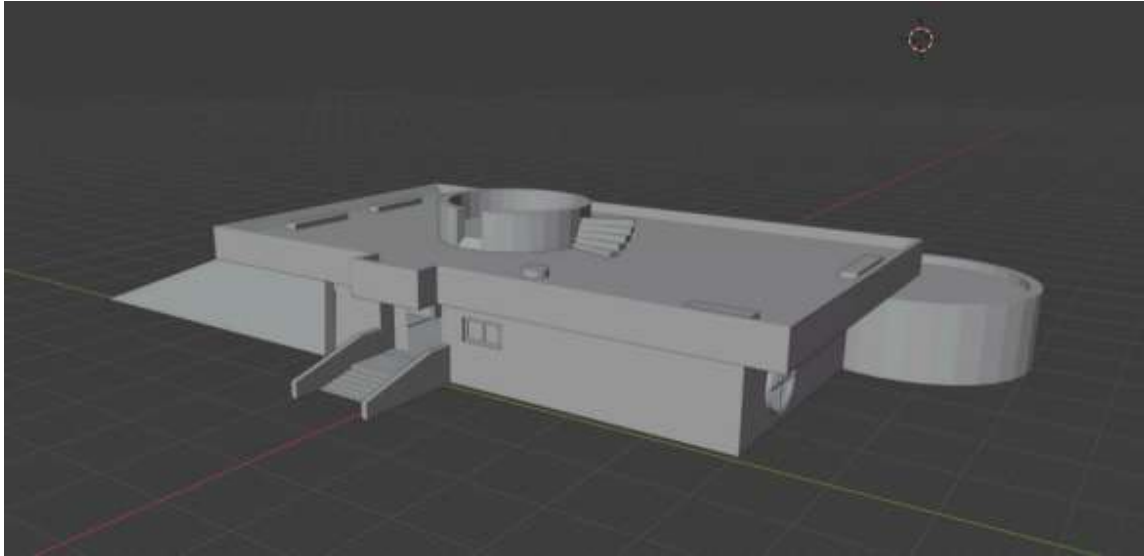


Image 129. Museum Modeling.

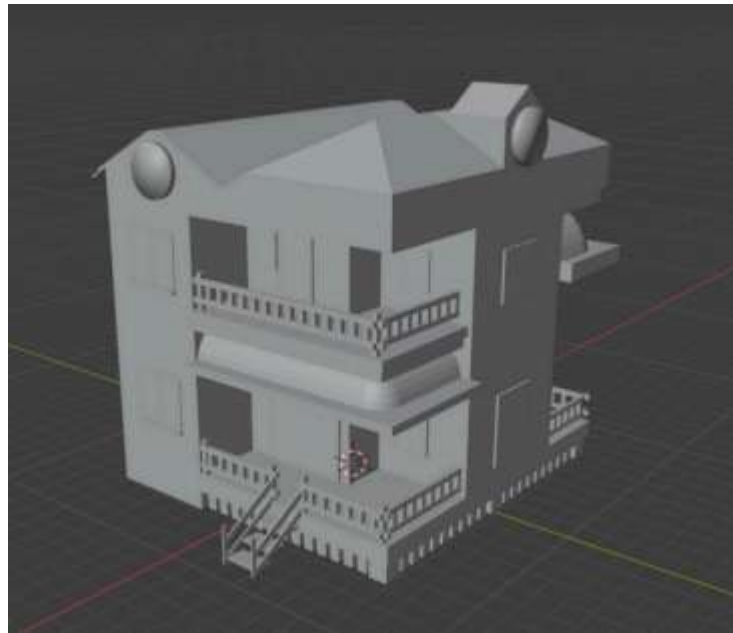


Image 130. House modeling.



Image 131. Eagle modeling.

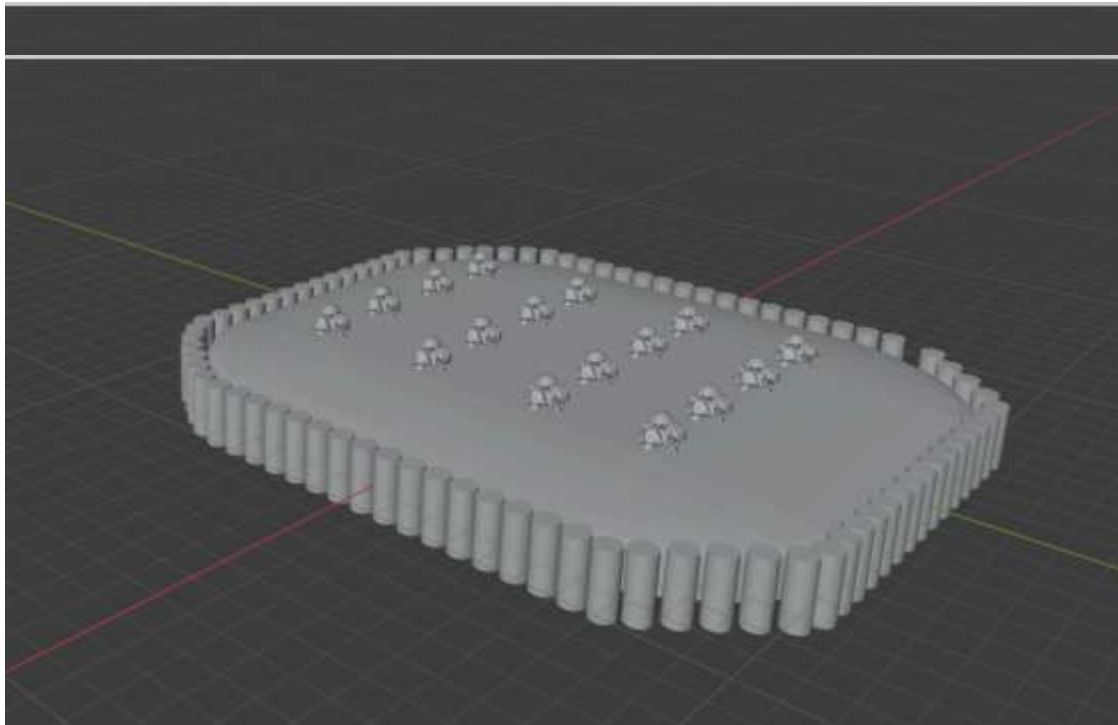


Image 132. Modeling of chinampas.

Textured:

For texturing, specific images were selected to represent the models' surfaces, such as stone, dirt, or wood. Thanks to the use of UV mapping, these textures could be applied directly to each model. After unwrapping, the image placement on each face of the object was manually adjusted, ensuring it was properly positioned, without distortion or pixelation.



Image 133. Sign texturing.

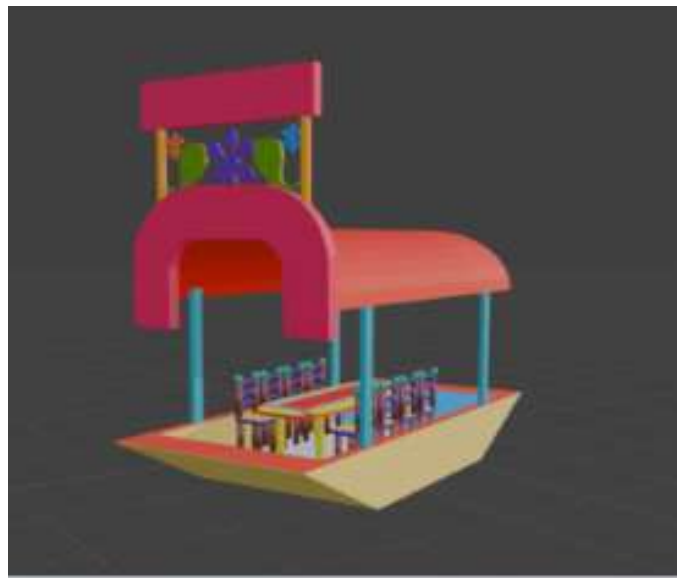


Image 134. Texturing of trajinera.

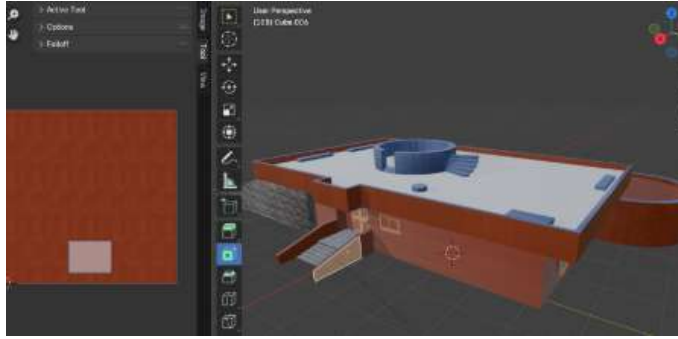


Image 135. Museum texturing.



Image 136. House texturing.



Image 137. Eagle texture.

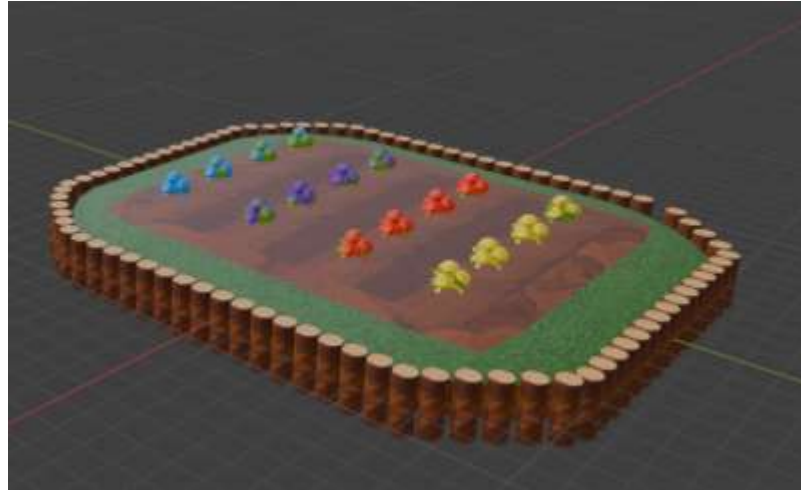


Image 138. Texture of chinampas.

Lightning:

The scene's lighting was implemented directly in OpenGL using shaders. The way light affects each surface was controlled, generating shadows, brightness, and contrasts that provide visual depth. This system allowed the park to be illuminated coherently and dynamically, highlighting materials, textures, and volumes in real time during the visualization.

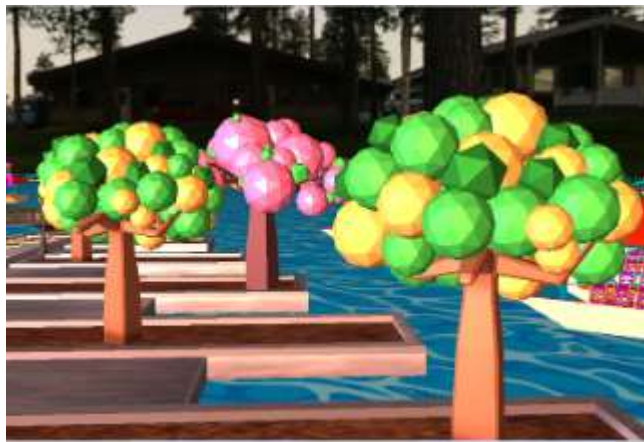


Image 139. Tree lighting.



Image 140. Bridge lighting.



Image 141. Museum lighting.



Image 142. Lighting of chinampas.



Image 143. Referring to model with lighting.



Image 144. Model referring to no lighting.

Materials:

Two approaches were combined in material representation. For some objects, textures were used that directly simulated the desired material, such as wood, water, or stone, so that they were immediately visually recognizable. In other cases, effects were applied using OpenGL code to modify visual properties such as gloss, transparency, or reflection, making the object react to light as the real material would.



Image 145. Wooden bench.



Image 146. We wash metal.

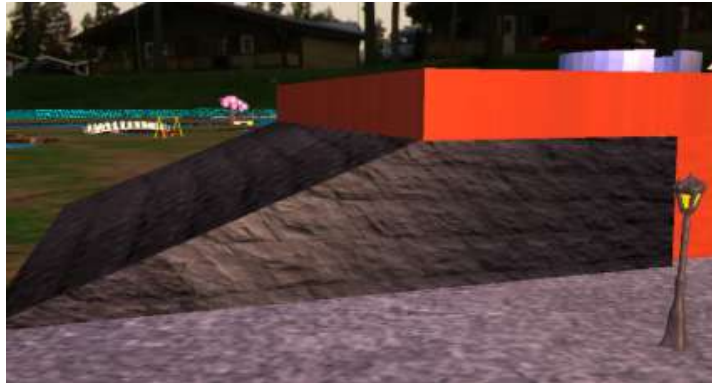


Image 147. Stone pyramid.

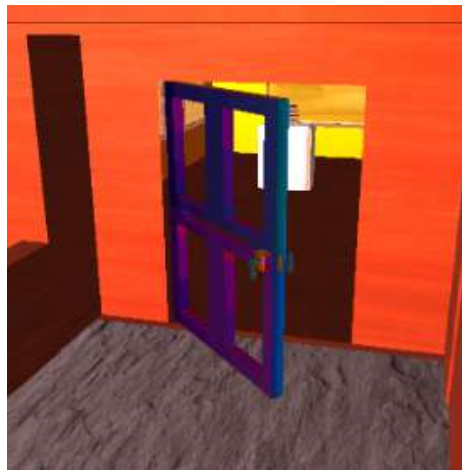


Image 148. Metal door.

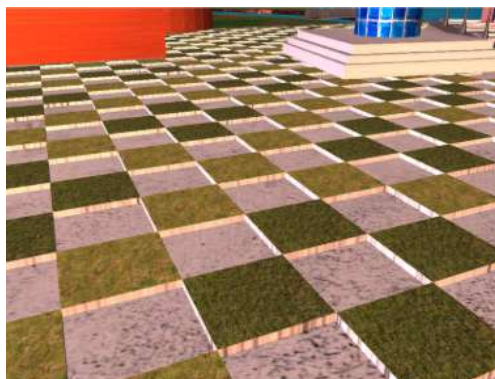


Image 149. Hail floor.

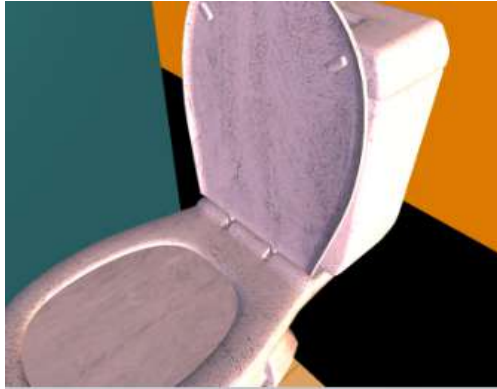


Image 150. WC.



Image 151. Park view.



Image 152. Rear view of the park.

Animation and Interaction:

For the animation of elements in the scene, an export in .fbx format was carried out. The positional manipulation of these objects was a crucial step to ensure their correct spatial integration. Specifically, the main museum door implements a rotation animation on its vertical axis, managed using GLM transformations. This interactive functionality allows users to open it with the H key and close it with the J key, directly affecting the `door_rotation` variable during the rendering process to control its real-time rotation.

```
model = glm::mat4(1.0f);  
model = glm::translate(model, door_position);  
model = glm::rotate(model, glm::radians(door_rotation), glm::vec3(0.0f, 1.0f, 0.0f));  
model = glm::rotate(model, glm::radians(-90.0f), glm::vec3(1.0f, 0.0f, 0.0f));  
mLightsShader->setMat4("model", model);  
door->Draw(*mLightsShader);
```

Image 153. Code used for museum door animation.

This code segment is fundamental for the manipulation and animation of 3D objects within the scene, specifically for the museum door. Initially, a model matrix is set to its identity value, serving as a starting point for all transformations. Subsequently, a translation is applied to position the object in its spatial location defined by `door_position`. Next, a rotation based on the `door_rotation` variable around the Y-axis is incorporated, which allows for the interactive opening and closing of the door according to user input. Additionally, a second fixed rotation of -90 degrees around the X-axis is applied, which is a common practice to adjust the initial orientation of the model and ensure its correct alignment within the application's coordinate system. Once all these transformations have been concatenated into the model matrix, it is sent to the light shader, `nLightsShader`, which will use it to process the object's geometry. Finally, the draw function of the door object is invoked, passing the already configured shader, resulting in the visual representation of the door in the scene with all its applied transformations.

```
if (glfwGetKey(window, GLFW_KEY_H) == GLFW_PRESS)
    door_rotation += 1.f;
if (glfwGetKey(window, GLFW_KEY_J) == GLFW_PRESS)
    door_rotation -= 1.f;
```

Image 154. Code used for museum door animation.

This code segment manages user interaction with the museum door. It uses the `glfwGetKey` function to detect the state of the H and J keys. When the H key is pressed, the `door_rotation` variable increases, initiating or continuing the door's opening movement. Conversely, if the J key is pressed, `door_rotation` decreases, causing the closing movement. This mechanism allows for direct, real-time control of the door's animation via keyboard input.

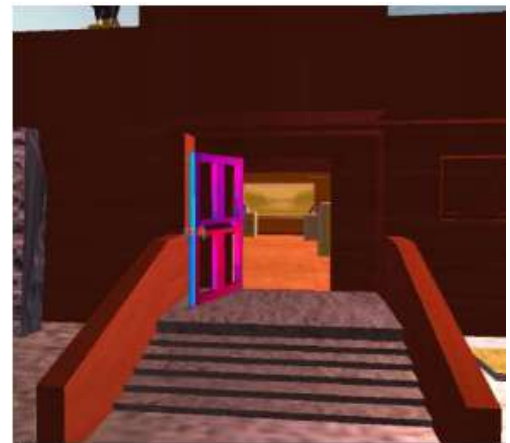


Image 155. For museum door animation.

The functionality of the fans in the virtual environment has been implemented using a continuous animation system, whose movement is governed by the `deltaTime` variable. This approach ensures that the rotation speed of both fans is consistent and independent of the system's frame rate, providing a fluid and uniform visual experience.

To offer user control over this animation, two specific keys have been assigned: the T key and the G key. Pressing the T key activates the rotation of the fans, initiating their uninterrupted movement. Conversely, pressing the G key completely stops the rotation, allowing the user to pause the animation according to their preference or need for interaction with the environment.

This design allows for dynamic management of animated elements, adding realism and interactivity to the project. The careful implementation of `deltaTime` guarantees that the rotation simulation is precise and adaptable to different hardware configurations, optimizing the user experience.

```
glm::mat4 model_fan = glm::mat4(1.0f);  
model_fan = glm::translate(model_fan, fan1_position);  
model_fan = glm::rotate(model_fan, glm::radians(-90.0f), glm::vec3(1.0f, 0.0f, 0.0f));  
model_fan = glm::rotate(model_fan, glm::radians(fan_rotation), glm::vec3(0.0f, 0.0f, 1.0f));  
  
mLightsShader->setMat4("model", model_fan);  
fan1->Draw(*mLightsShader);
```

Image 156. Code for fan movement.

This code segment is responsible for positioning and animating the 3D model of a fan. The model matrix is initialized and then a translation is applied to place the fan at `fan1_position`. Next, two rotations are performed: a fixed rotation of -90 degrees around the X-axis to correct the model's orientation, and a dynamic rotation around the Z-axis, controlled by the `fan_rotation` variable, which simulates the continuous rotation of the fan. Finally, the resulting model matrix is passed to the light shader, `nLightsShader`, so that the fan is rendered correctly in the scene.

```
if (glfwGetKey(window, GLFW_KEY_T) == GLFW_PRESS) {  
    fan_rotate = true;  
    fan2_rotate = true;  
}  
  
if (glfwGetKey(window, GLFW_KEY_G) == GLFW_PRESS) {  
    fan_rotate = false;  
    fan2_rotate = false;  
}
```

Image 157. Code for fan movement.

This code snippet controls the start and stop of the fans' animation. When the user presses the T key, the boolean variables `fan_rotate` and `fan2_rotate` are set to true. This activates the continuous rotation mechanism for both fans.

On the other hand, if the user presses the G key, both `fan_rotate` and `fan2_rotate` variables are set to false. This stops the fans' rotation, pausing their animation. This code provides a simple interface for the user to control the animation state of the fans in real time.



Image 158. Code for fan movement.

The trajinera's animation has been implemented with an interactive approach that responds to user proximity within the virtual environment. This characteristic Xochimilco boat begins its rotation automatically and continuously when the user is less than 10 units away from it. This proximity mechanism is managed by a constant check of the distance between the user's current position and the trajinera's location. As long as the user remains within this 10-unit radius, the `trajinera_rotation` variable progressively increases in each rendering cycle. This constant increment of the `trajinera_rotation` variable directly translates into a continuous rotation of the trajinera on its own axis, creating a sense of movement and life within the virtual ecosystem.

The objective of this interaction is twofold: on one hand, it adds a dynamic and reactive element to the scenery, making exploration more immersive; on the other hand, it can serve as a visual signal or an incentive for the user to approach and explore this culturally significant element of the Xochimilco Ecological Park more closely. The rotation automatically ceases once the user moves beyond the 10-unit threshold, maintaining contextual control of the animation.

```
float dist_trajinera = glm::distance(camera.Position, trajinera_position);  
  
if (dist_trajinera < 10.0f) {  
    trajinera_rotation += 60.0f * deltaTime;  
    if (trajinera_rotation > 360.0f) trajinera_rotation -= 360.0f;  
}
```

Image 159. Code for trajinera animation.

This code block implements the logic for the trajinera's interactive rotation. First, it calculates the distance between the current camera position (representing the user) and the trajinera's position using `glm::distance`. If this `dist_trajinera` is less than 10.0 units, the `trajinera_rotation` variable is incremented by 60.0 degrees multiplied by `deltaTime`. The use of `deltaTime` ensures that the rotation speed remains constant regardless of the system's frame rate. Finally, a condition is included so that `trajinera_rotation` resets to 0.0 degrees once it exceeds 360.0 degrees, keeping the angle within a valid circular range and preventing the variable from growing indefinitely.

```
glm::mat4 model_trajinera = glm::mat4(1.0f);
model_trajinera = glm::translate(model_trajinera, trajinera_position);
model_trajinera = glm::rotate(model_trajinera, glm::radians(trajinera_rotation), glm::vec3(0.0f, 1.0f, 0.0f));
model_trajinera = glm::rotate(model_trajinera, glm::radians(-90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
nLightsShader->setMat4("model", model_trajinera);
trajinera->Draw(*nLightsShader);
```

Image 160. Code for trajinera animation.

This code segment is responsible for applying model transformations to the trajinera for its correct positioning and animation in the scene. A model matrix for the trajinera (`model_trajinera`) is initialized with an identity matrix. Next, the model is translated to its specific location defined by `trajinera_position`. Then, two rotations are applied: the first uses `trajinera_rotation` to rotate the trajinera around the Y-axis, which allows for its interactive animation based on user proximity. The second is a fixed rotation of -90 degrees around the X-axis, used to ensure the correct orientation of the model. Finally, the `model_trajinera` matrix, which already contains all accumulated transformations, is sent to the light shader (`nLightsShader`), and the trajinera model is then drawn on the screen with the shader applied.



Image 161. Trajinera animation.

The appearance of the ghost in the scene is a dynamic effect that is conditionally activated, emerging over the book only when the boolean variable `show_book_info` is set to true. This functionality aims to add an interactive and narrative element to the environment. The ghost's upward movement is controlled by progressively increasing the `popup_height` variable, an operation synchronized with time (`deltaTime`) to ensure a fluid and consistent animation, regardless of the system's frame rate.

Once the ghost begins its ascent, its model is transformed through a series of operations. First, a translation is applied along the Y-axis based on `popup_height` to simulate its elevation from the book. Subsequently, rotations are performed to adjust its orientation and a scaling is applied to define its final size, ensuring that the model is displayed correctly and with the desired visual impact on the screen.

```

if (show_book_info) {
    popup_height += deltaTime * 1.5f;
    if (popup_height > 1.5f) popup_height = 1.5f;

    glm::vec3 ghost_pos = book_position + glm::vec3(0.0f, popup_height + 1.0f, 0.0f);
    glm::mat4 ghost_model_mat = glm::mat4(1.0f);
    ghost_model_mat = glm::translate(ghost_model_mat, ghost_pos);
    ghost_model_mat = glm::rotate(ghost_model_mat, glm::radians(-90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
    ghost_model_mat = glm::scale(ghost_model_mat, glm::vec3(0.45f));

    mLightsShader->setMat4("model", ghost_model_mat);
    ghost_model->Draw(*mLightsShader);
}

```

Image 162. Code for ghost animation.

This code block controls the logic for the ghost model's appearance and transformation, which is activated only if the `show_book_info` variable is true. Within this condition, `popup_height` progressively increases by 1.5 units per `deltaTime`, simulating the ghost's upward movement. Once `popup_height` reaches a value of 1.5, it is capped at that value to stop the ascent. Subsequently, the ghost's final position (`ghost_pos`) is calculated by adding the book's position (`book_position`) to a vector that includes `popup_height` and an additional vertical offset. Subsequently, the ghost's final position (`ghost_pos`) is calculated by adding the book's position (`book_position`) to a vector that includes `popup_height` and an additional vertical offset.



Image 163. Ghost animation.

Additionally, the bathroom door in our virtual environment has been designed to offer a fluid and natural interactive experience to the user. It incorporates a proximity detection system that automates its opening and closing, eliminating the need for direct manual interaction. The operation is simple yet effective: the door opens automatically the moment the user approaches within a distance of less than 10 units. This distance threshold is continuously calculated in each game frame, using the `glm::distance` function, which measures the exact separation between the user's position (represented by the camera) and the center of the door. Once the user crosses this limit, the door begins its opening movement.

Conversely, if the user moves away and exceeds 10 units of distance, the door detects this change and begins to close by itself. This entire opening and closing process is achieved by progressively adjusting the `door_bath_rotation` variable in each frame. The key to a smooth and consistent animation, regardless of team performance, lies in the use of `deltaTime`. This value ensures that the door's rotation speed is uniform, providing a visually pleasing and realistic transition that contributes to the user's immersion in the museum environment.

```
float dist = glm::distance(camera.Position, door_bath_position);

// Abrir si está cerca
if (dist < 10.0f && door_bath_rotation < 100.0f) {
    door_bath_rotation += 60.0f * deltaTime;
}

// Cerrar si ya se alejó
else if (dist >= 10.0f && door_bath_rotation > 0.0f) {
    door_bath_rotation -= 60.0f * deltaTime;
}
```

Image 164. Code for bathroom door animation.

This code snippet manages the logic for the automatic opening and closing of the bathroom door based on user proximity. First, the distance (`dist`) between the camera's current position (the user) and the bathroom door's position (`door_bath_position`) is calculated using the `glm::distance` function. Next, two

main conditions are evaluated. If the distance (dist) is less than 10.0 units and the door's rotation angle (door_bath_rotation) is less than 100.0 degrees (indicating the door is not fully open), then door_bath_rotation is incremented by 60.0 degrees multiplied by deltaTime. This causes the door to open gradually.

Conversely, if the distance (dist) is equal to or greater than 10.0 units (meaning the user has moved away) and the door's rotation angle (door_bath_rotation) is greater than 0.0 degrees (indicating the door is not fully closed), then door_bath_rotation is decremented by 60.0 degrees multiplied by deltaTime. This action progressively closes the door. The use of deltaTime in both cases ensures that the opening and closing speed is consistent and fluid, regardless of the system's frame rate.

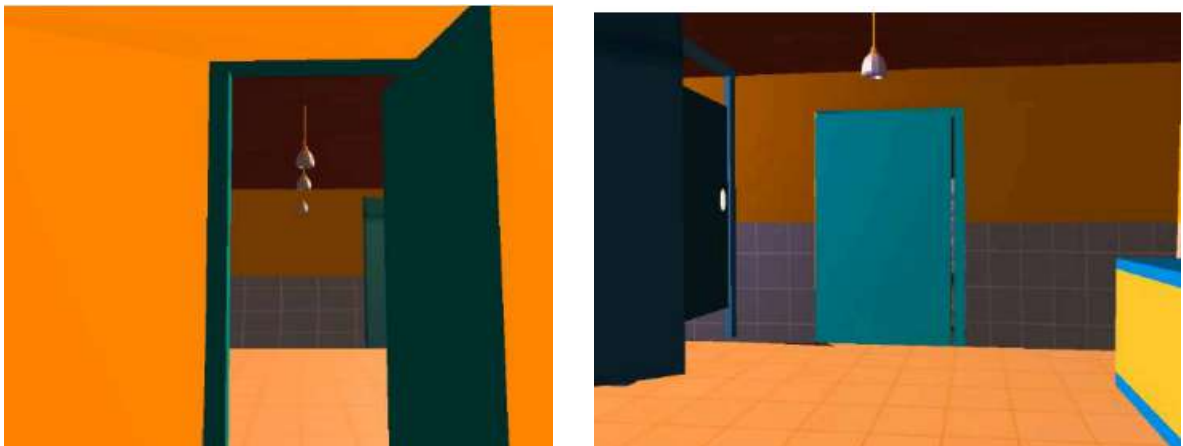


Image 165. Animation for the bathroom door.

Subsequently, the lake surface is animated using a procedural shader, a technique that dynamically generates visual effects on the GPU. This allows for simulating realistic wave motion, creating the illusion that the water is moving with authentic real-time waves. The effect is achieved by continuously incrementing a time variable, `wavesTime`, in each frame. This vertex manipulation produces the effect of waves that propagate and change over time.

```
wavesShader->setFloat("time", wavesTime);  
wavesShader->setFloat("radius", 5.0f);  
wavesShader->setFloat("height", 5.0f);  
  
gridMesh->Draw(*wavesShader);  
wavesTime += 0.0003;
```

Image 166. Code for lake surface.

This code snippet is responsible for configuring the shader parameters for water simulation and controlling wave animation. It begins by sending the `wavesTime` variable to the `wavesShader` as a uniform named "time." This constant increment of `wavesTime` in each frame is crucial for the shader to generate continuous and fluid wave motion, giving the illusion that the water is constantly agitated.



Image 167. Animation for the lake surface.

Final results:

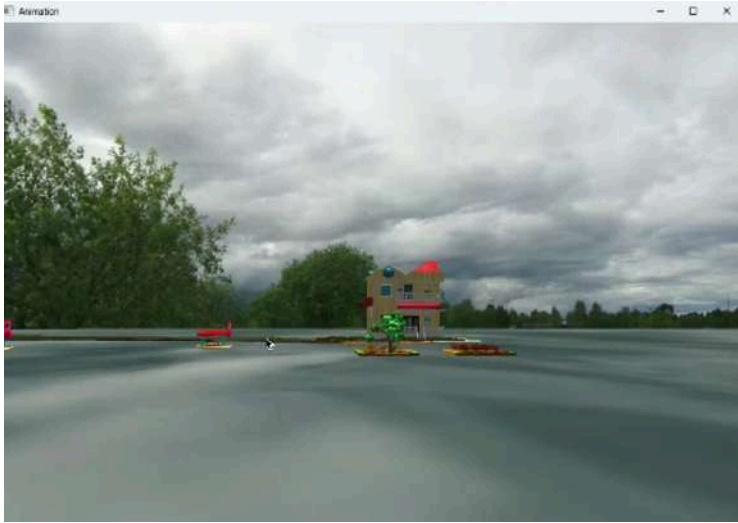


Image 168. View in OpenGL



Image 169. View in OpenGL

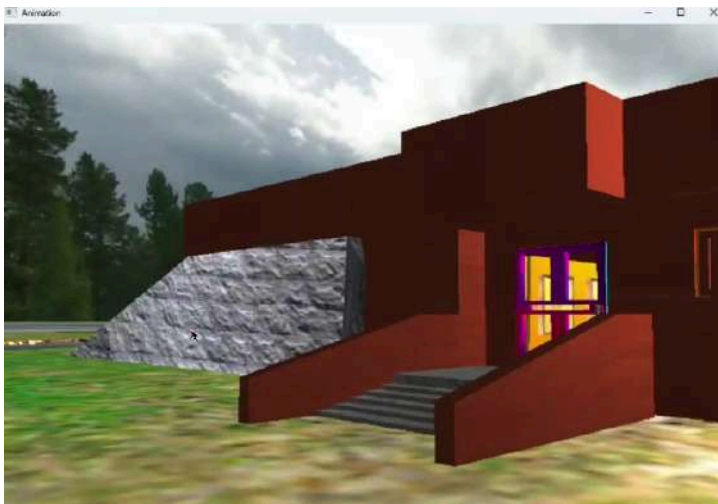


Image 170. View in OpenGL



Image 171. View in OpenGL



Image 172. View in OpenGL



Image 173. View in OpenGL.

INDIVIDUAL CONCLUSIONS

318245537:

This project has been a significant milestone in my journey as an engineering student. Not only did I have the opportunity to delve into the intricate world of graphic application development, but I also solidified and expanded my knowledge of OpenGL, complex hierarchical structures, and precise animation control. The staged planning methodology proved to be fundamental, acting as a compass that guided us through the various phases of the project. This strategy allowed us to maintain rigorous order and advance consistently, especially when distributing tasks among team members, assigning responsibilities based on each individual's strengths.

One of the most valuable lessons learned was understanding firsthand how collaborative work, built upon a clear and shared ecological vision, can transform into a powerful tool for shaping an interactive experience that goes

beyond mere entertainment. This experience showed me that technology can be a bridge to connect people with pressing environmental issues. However, I'm aware that there are still opportunities to further enrich this project. I envision incorporating a more immersive sound system, capable of enveloping the user in the Xochimilco environment, or implementing a decision-making system that allows the user to influence the course of the virtual tour. These additions could significantly increase the project's emotional and educational impact.

Despite the inherent technical challenges of a project of this magnitude, this experience only reaffirmed my deep interest in dedicating my career to combining engineering with social and environmental causes. I firmly believe in the potential of technological solutions to generate positive change in our environment and society. This project not only provided me with valuable technical skills but also inspired me to continue exploring how technology can be a force for good.

318069434:

The completion of this project has been a deeply revealing experience, allowing me to tangibly understand how a meticulously structured methodology based on detailed planning, hierarchical modeling, and expert OpenGL implementation directly and significantly impacts the final quality of any project. Throughout this journey, which spanned from the initial conception of the idea to the final integration of each component, I personally experienced the invaluable worth of breaking down intricate tasks into manageable, well defined stages. This approach not only optimized the workflow but also gave us the opportunity to apply both our accumulated technical knowledge and our emerging creative skills in a synergistic and effective way. This methodology, far beyond being a mere organizational outline, proved to be a powerful tool for team management and strategic decision making. It enabled us to coordinate efficiently, facilitating the choice of animation techniques that best suited our vision, the application of

realistic lighting effects that enriched the atmosphere of the virtual environment, and ensuring that the final product was not only functional but also educational and capable of conveying the desired message. Collaborative work, guided by a precise timeline and intelligent role distribution, turned out to be key to overcoming complex challenges. From optimizing 3D models for optimal performance to the impeccable synchronization of animations and the final packaging of the product, every step was a testament to the power of teamwork and effective communication.

Ultimately, this project has been much more than a technical exercise. It has been a formative experience that has not only strengthened my technical skills in the field of computer graphics but has also reaffirmed my personal commitment to the design and development of technological solutions that actively promote environmental awareness and sustainability. I feel inspired and motivated to continue exploring how technology can be a force for good, helping to create a more conscious and harmonious future for our planet.

418046210:

This project has been much more than just the creation of a graphic application. It represents a significant intersection between computer graphics technology and a deep concern for environmental and cultural sustainability. By immersing ourselves in the task of virtually recreating the Xochimilco Ecological Park, we not only set out to demonstrate our technical skills in 3D modeling, animation, and OpenGL programming, but we also embarked on a journey of learning and discovery about the richness and fragility of this unique ecosystem.

Through the development of this interactive environment, we have sought to offer a platform that allows users to experience the beauty of Xochimilco firsthand, but also to understand the pressing challenges it faces. The interactivity of the virtual environment facilitates the exploration of its canals,

chinampas, and the diversity of flora and fauna, creating a personal and emotional connection with the place. We hope this virtual experience encourages reflection and active commitment to conservation. By allowing users to virtually "walk" through this space, learn about its pollution problems, and the importance of ecological restoration, we aim to plant a seed of environmental awareness that transcends the screen.

Beyond programming and design, this project has taught us about the importance of interdisciplinary collaboration. Integrating knowledge about biology, ecology, and local culture was fundamental to creating an authentic and educational representation of Xochimilco. Every 3D model, every texture, and every animation has been informed by research and learning about this site. Furthermore, teamwork has been essential; coordination, task division, and constant communication allowed us to overcome technical and conceptual obstacles.

321589163:

The development of this graphics project turned out to be a remarkably complex task that encompassed far more than the technical concepts we are familiar with, such as mathematics, geometry, linear algebra, and vector calculus. It also extended into the field of graphic arts, especially in the processes of modeling, texturing, and animation. It quickly became evident that this was not solely an engineering project, but also an artistic endeavor, which gave it a truly multidisciplinary character.

Personally, I experienced greater difficulty with the modeling and texturing aspects, which demanded an attention to detail and an ability to visualize forms in three dimensions that I had not anticipated. The loading of models also presented a considerable challenge, as we were forced to optimize some geometries so that the program could run smoothly on as many computers as

possible. This aspect proved to be a crucial point where programming and art intertwined, demanding a delicate balance between technical efficiency and visual quality. At the program level, we used a series of tools with which we have become familiar in the lab, such as shaders, and projection, translation, rotation, and scaling matrices, to handle the camera, interact with the scene, and visualize the various animations. The application of these concepts in a practical context proved to be an enriching experience and strengthened my understanding of their function.

Therefore, I can affirm that this has probably been the most complex and largest project I have worked on since I entered the Faculty of Engineering. The scale of the work, as well as the different disciplines it covered, made it a significant challenge, but also an invaluable learning opportunity. This project taught me fundamental concepts that are used in even more complex applications, such as simulation software, graphics engines, and computer aided design software. I am sure that the skills and knowledge I acquired in this process will be of great use to me in my future as an engineer.

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